SDS 4030 - Statistics for Data Science II: Final Project

Main Markdown File

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Preliminary library loading, reading code in, and exploratory data analysis

```r
gss = read.table("gss.txt", header = TRUE)
summary(gss)
```

```
##       Age             Educ           Hours           Sex
##  Min.   :18.00   Min.   : 4.00   Min.   : 3.00   Length:815
##  1st Qu.:32.00   1st Qu.:12.00   1st Qu.:39.00   Class :character
##  Median :42.00   Median :14.00   Median :40.00   Mode  :character
##  Mean   :42.91   Mean   :13.93   Mean   :42.59
##  3rd Qu.:52.00   3rd Qu.:16.00   3rd Qu.:50.00
##  Max.   :88.00   Max.   :20.00   Max.   :89.00
##     TVHours          Happy               MEduc           FEduc
##  Min.   : 0.000   Length:815         Min.   : 0.00   Min.   : 0.00
##  1st Qu.: 2.000   Class :character   1st Qu.:10.00   1st Qu.: 9.00
##  Median : 2.000   Mode  :character   Median :12.00   Median :12.00
##  Mean   : 2.709                      Mean   :11.61   Mean   :11.45
##  3rd Qu.: 3.000                      3rd Qu.:14.00   3rd Qu.:14.00
##  Max.   :20.000                      Max.   :20.00   Max.   :20.00
##    Marital               Sibs            Work              Zodiac
##  Length:815         Min.   : 0.000   Length:815         Length:815
##  Class :character   1st Qu.: 2.000   Class :character   Class :character
##  Mode  :character   Median : 3.000   Mode  :character   Mode  :character
##                     Mean   : 3.439
##                     3rd Qu.: 5.000
##                     Max.   :16.000
##      Income
##  Min.   :     500
##  1st Qu.:   23750
##  Median :   45000
##  Mean   :  324767
##  3rd Qu.:   67500
##  Max.   :28637626
```

We begin our data science workflow by familiarizing ourselves with the GSS dataset. The dataset contains 815 rows corresponding to 815 individuals' survery responses. The dataset contains 13 variables. There are 8 discrete numerical variables: Age (Responder's Age), Educ (Total years of education), Hours (Weekly work hours), TVHours (Hours spent watching TV daily), MEduc (Mother's

years of education), FEduc (Father's years of education), Sibs (Number of siblings), and Income (Yearly income). Thus there are 5 categorical variables: Sex (Responder's sex), Happy (Happiness level), Marital (Marital status), Work (Work status), and Zodiac (Zodiac sign). Uniquely they take on the following values: Sex (Female, Male), Happy (Pretty Happy, Very Happy, Not Too Happy), Marital (Divorced, Widowed, Married, Never Married, Separated), Work (Fulltime, Housekeeping, Laid Off, School, Retired, Parttime, Not Working, Other), and Zodiac (Virgo, Libra, Taurus, Pisces, Cancer, Aquarius, Leo, Capricorn, Aries, Gemini, Sagittarius, Scorpio).

Now that we understand our data we isolate our goal being to build a linear model for the response variable yearly income by treating all other seven quanitative variables as covariates and ignoring all five qualitative variables.

Step 1:

i) Fit a multiple linear model with all available quantitative covariates.

ii) Perform exact and asymptotic tests for the set of linear constraints which has been specified for your team.

iii) Do all of your conclusions agree with each other and what are the relationships between the different tests?

---

i) Fitting a multiple linear model with all available quantitative covariates. This is the model that we will use for the next few parts of this project

```r
library(faraway)
full_model <- lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc +
    Sibs, gss)
summary(full_model)
```

```
##
## Call:
## lm(formula = Income ~ Age + Educ + Hours + TVHours + MEduc +
##     FEduc + Sibs, data = gss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1211280  -410167  -271382  -129856 28149890
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   211943     688636   0.308   0.7583
```

```
## Age             2949       6231   0.473   0.6361
## Educ           20448      35626   0.574   0.5661
## Hours          -9324       5923  -1.574   0.1159
## TVHours       -65363      42493  -1.538   0.1244
## MEduc          -3715      32699  -0.114   0.9096
## FEduc          10206      28724   0.355   0.7225
## Sibs           58649      33490   1.751   0.0803 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2340000 on 807 degrees of freedom
## Multiple R-squared:  0.0101, Adjusted R-squared:  0.001516
## F-statistic: 1.177 on 7 and 807 DF,  p-value: 0.3136
```

For the sake of later model comparison we will find the least squares estimate of the coefficient vector, an unbiased estimate of the residual variance and the MLE of the residual variance in the full model.

```r
Y = gss$Income
X = model.matrix(full_model)
n = dim(X)[1]
p = dim(X)[2]
beta = solve(crossprod(X), crossprod(X, Y))[, ]
print(beta)
```

```
## (Intercept)         Age        Educ       Hours     TVHours       MEduc
##  211942.841    2949.415   20448.463   -9323.480  -65363.112   -3715.210
##       FEduc        Sibs
##   10205.481   58648.969
```

```r
residuals = Y - X %*% beta
sigma = sqrt(mean(residuals^2))
print(sigma^2)
```

```
## [1] 5.42076e+12
```

```r
S = sqrt(sum(residuals^2)/(n - p))
print(S^2)
```

```
## [1] 5.474497e+12
```

Our full model is $Y_i = \beta_0 + \beta_{Age}\text{Age}_i + \beta_{Educ}\text{Educ}_i + \beta_{Hours}\text{Hours}_i + \beta_{TVHours}\text{TVHours}_i + \beta_{MEduc}\text{MEduc}_i + \beta_{FEduc}\text{FEduc}_i + \beta_{Sibs}\text{Sibs}_i + \epsilon_i$. An unbiased estimate of the residual variance is equal to 5.42076e+12 and the MLE of the residual variance is equal to 5.474497e+12, under the full model.

The reduced model is a little more tricky where we have to create a model in which it follows the constraints given by: $H_0 : 10 * \beta_{MEduc} = 20 * \beta_{FEduc} = 2$.

We can rewrite the constraint as $10 * \beta_{MEduc} = 2$, $20 * \beta_{FEduc} = 2$. Or, $\beta_{MEduc} = 0.2$, $\beta_{FEduc} = 0.1$

We can plug the null values into the full model, replacing coefficients with the constants so the full model becomes $Y_i = \beta_0 + \beta_{Age}\text{Age}_i + \beta_{Educ}\text{Educ}_i + \beta_{Hours}\text{Hours}_i + \beta_{TVHours}\text{TVHours}_i + 0.2 * \text{MEduc}_i + 0.1 * \text{FEduc}_i + \beta_{Sibs}\text{Sibs}_i + \epsilon_i$

We have to set MEduc and FEduc, the fixed parts of the response in the reduced model, as the offset.

```r
reduced_model <- lm(Income ~ Age + Educ + Hours + TVHours + Sibs, gss,
    offset = 0.2 * MEduc + 0.1 * FEduc)
summary(reduced_model)
```

```
##
## Call:
## lm(formula = Income ~ Age + Educ + Hours + TVHours + Sibs, data = gss,
##     offset = 0.2 * MEduc + 0.1 * FEduc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1208997  -397497  -267841  -139546 28119340
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   286038     616017   0.464   0.6425
## Age             2334       5805   0.402   0.6878
## Educ           23374      33141   0.705   0.4808
## Hours          -9411       5911  -1.592   0.1117
## TVHours       -66218      42346  -1.564   0.1183
## Sibs           56138      32385   1.733   0.0834 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2337000 on 809 degrees of freedom
## Multiple R-squared:  0.009939,   Adjusted R-squared:  0.00382
## F-statistic: 1.624 on 5 and 809 DF,  p-value: 0.151
```

```r
R <- rbind(c(0, 0, 0, 0, 0, 1, 0, 0), c(0, 0, 0, 0, 0, 0, 1, 0))
# the 1's in R represent the MEduc and FEduc since those are within
# the null hypothesis
r <- c(0.2, 0.1)
k <- length(r)
beta0 = beta - solve(crossprod(X), crossprod(R, solve(R %*% solve(crossprod(X),
    t(R)), R %*% beta - r)))[, ]
print(beta0)
```

```
## (Intercept)         Age        Educ       Hours     TVHours       MEduc
##  286037.934    2333.632   23374.285   -9411.360  -66218.159       0.200
##       FEduc        Sibs
##       0.100   56138.450
```

```
residuals0 = Y - X %*% beta0
sigma0 = sqrt(mean(residuals0^2))
print(sigma0^2)
```

```
## [1] 5.421657e+12
```

```
S0 = sqrt(sum(residuals0^2)/(n - p + k))
print(S0^2)
```

```
## [1] 5.461868e+12
```

An unbiased estimate of the residual variance is equal to 5.421657e+12 and the MLE of the residual variance is equal to 5.461868e+12, under the reduced model.

ii) Perform exact and asymptotic tests for the set of linear constraints which has been specified for your team.

We have to test the hypothesis $H_0 : 10 * \beta_{MEduc} = 20 * \beta_{FEduc} = 2$ against exact and asymptotic tests. The best exact test is, of course, the exact F-test assuming we are have normalcy conditions (which we will check later, but step 1 ensures that we do it regardless of the normalcy checks). To test for the asymptotic tests it is logical to use the Wald test since we have a large sample (n = 815). It is also one of the easier asymptotic tests to check, so we will use it due to its efficiency, accuracy, and sample size.

Exact F- Test

```
FT = crossprod(R %*% beta - r, solve(R %*% solve(crossprod(X), t(R)),
    R %*% beta - r))[, ]/(k * S^2)
print(FT)
```

```
## [1] 0.06680263
```

```
pvalue = pf(FT, k, n - p, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.935385
```

```r
anova(reduced_model, full_model)
```

```
## Analysis of Variance Table
##
## Model 1: Income ~ Age + Educ + Hours + TVHours + Sibs
## Model 2: Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs
##   Res.Df        RSS Df  Sum of Sq      F Pr(>F)
## 1    809 4.4187e+15
## 2    807 4.4179e+15  2 7.3142e+11 0.0668 0.9354
```

The observed value of the F test statistic is equal to 0.06680263 and the corresponding p-value is equal to 0.935385, so we fail to reject the null hypothesis at significance level $\alpha = 5\%$.

Wald Test

```r
WT = crossprod(R %*% beta - r, solve(R %*% solve(crossprod(X), t(R)),
    R %*% beta - r))[, ]/S^2
print(WT)
```

```
## [1] 0.1336053
```

```r
pvalue = pchisq(WT, k, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.9353798
```

```r
anova(reduced_model, full_model, test = "Chisq")
```

```
## Analysis of Variance Table
##
## Model 1: Income ~ Age + Educ + Hours + TVHours + Sibs
## Model 2: Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs
##   Res.Df        RSS Df  Sum of Sq Pr(>Chi)
## 1    809 4.4187e+15
## 2    807 4.4179e+15  2 7.3142e+11   0.9354
```

The observed value of the Wald test statistic is equal to 0.1336053 and the corresponding p-value is equal to 0.9353798, so we fail to reject the null hypothesis at significance level $\alpha = 5\%$

Score Test

```r
SF = crossprod(X, residuals0)/sigma0^2
FI = crossprod(X)/sigma0^2
ST = crossprod(SF, solve(FI, SF))[, ]
print(ST)
```

```
## [1] 0.1349074
```

```r
P = X %*% solve(crossprod(X), t(X))
ST = sum((P %*% residuals0)^2)/sigma0^2
print(ST)
```

```
## [1] 0.1349074
```

```r
pvalue = pchisq(ST, k, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.934771
```

The observed value of the Score test statistic is equal to 0.1349074 and the corresponding p-value is equal to 0.934771, so we fail to reject the null hypothesis at significance level $\alpha = 5\%$.

Likelihood Ratio Test

```r
LR = -2 * (logLik(reduced_model)[1] - logLik(full_model)[1])
print(LR)
```

```
## [1] 0.1349186
```

```r
LR = n * log(sigma0^2/sigma^2)
print(LR)
```

```
## [1] 0.1349186
```

```r
pchisq(LR, k, lower.tail = FALSE)
```

```
## [1] 0.9347658
```

```r
library(lmtest)
lrtest(reduced_model, full_model)
```

```
## Likelihood ratio test
##
## Model 1: Income ~ Age + Educ + Hours + TVHours + Sibs
## Model 2: Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs
##   #Df LogLik Df  Chisq Pr(>Chisq)
## 1   7 -13105
## 2   9 -13105  2 0.1349     0.9348
```

The observed value of the Likelihood Ratio test statistic is equal to 0.1349186 and the corresponding p-value is equal to 0.9347658, so we fail to reject the null hypothesis at significance level $\alpha = 5\%$.

iii) Do all of your conclusions agree with each other and what are the relationships between the different tests?

Yes, all of my conclusions agree with each other as in the exact F test, Wald test, score test, and likelihood ratio test we have failed to reject the null, and they all yield similar p-values. Both the F-Exact test and asymptotic tests provide very small test statistics which means we do not have substantial evidence to reject the null hypothesis.

```r
WT = k * FT
print(WT)
```

```
## [1] 0.1336053
```

```r
ST = WT * S^2/sigma0^2
print(ST)
```

```
## [1] 0.1349074
```

```r
LR = n * log(n/(n - p) * WT/ST)
print(LR)
```

```
## [1] 0.1349186
```

```r
LR = n * log(1 + WT/(n - p))
print(LR)
```

```
## [1] 0.1349186
```

```r
LR = -n * log(1 - ST/n)
print(LR)
```

```
## [1] 0.1349186
```

In the F test, the exact F-statistic is 0.06680263, and the p-value is 0.935385. In the Wald test the Wald statistic is 0.1336053 and the p-value is 0.9353798. According to Slutsky's theorem, we infer that W_n = kF, the Wald statistic equals kF asymptotically. The Wald statistic is k times the 0.06680263 where k=2. While the F test relies on a finite sample, the Wald statistic follows a chi-square distribution and asymptotic approximations.

In the score test, the score test statistic is 0.1349074 and the p-value is 0.934771 We know that $\hat{\sigma}_0^2 ST = S^2 W_n$. The Wald and Score test statistics only differ in terms of the residual variance estimation. The Wald test statistic utilizes an unbiased estimator of the residual variance under the full model while the Score test statistic utilizes the MLE of the residual variance under the reduced model. In other words the Wald test uses the full model variance and the Score test uses reduced model variance. Thus we can see how the two test statistics differ slightly.

In the likelihood ratio test, likelihood test statistic is 0.1349186 and the p-value is 0.9347658. Below we see how the Likelihood test statistic relates to the Wald test and Score test through transformations, implying they provide the same inference for very large sample sizes.

So yes, we observe that the results of all the four test statistics concur. The results of the asymptotic Wald, Likelihood Ratio and Score tests are perfectly in line with the results of the exact F test, which is expected since n = 815 which is large enough to be accurate for an asymptotic test. All three asymptotic tests, Wald, Score & Likelihood Ratio Tests, are all asymptotically equivalent, converging to the same test statistics and giving us the same conclusions. They measure deviation from the null differently, where the Wald test measures the horizontal differences of the parameters and the null, the Score test measures the slope of the log likelihood at the null, and the likelihood ratio test measures the vertical differences in log likelihood.

Step 2:

i) Is the variance of the random error terms non-constant as a function of any of the quantitative or qualitative covariates?

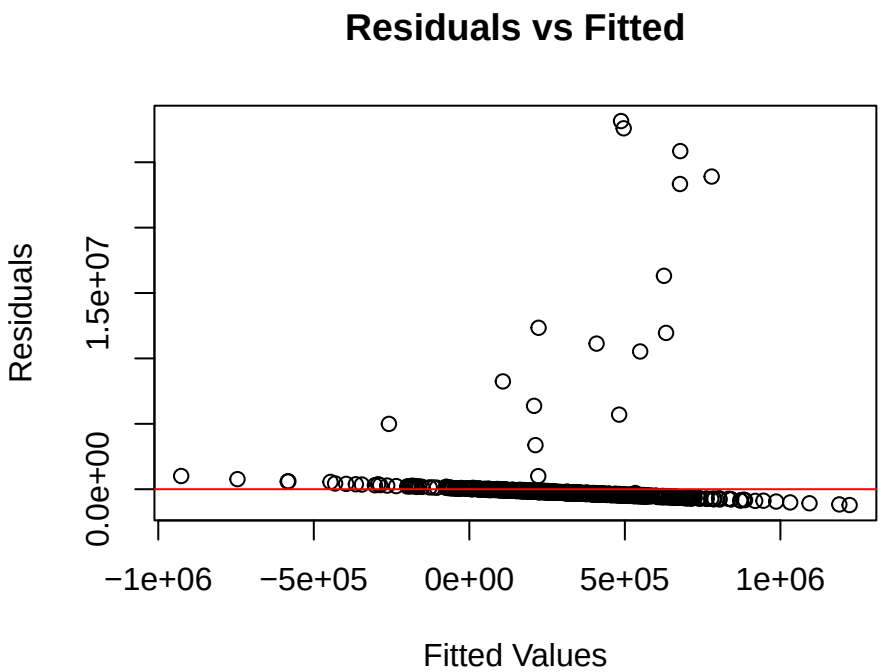
ii) Perform suitable heteroscedasticity tests and state your conclusions.

---

i) Is the variance of the random error terms non-constant as a function of any of the quantitative or qualitative covariates?

We begin by plotting and examining a residuals vs fitted plot of the full model (shown in part 1). We can note that the residuals increase as the fitted values increases and also has some extreme outliers and leverage points (also called influential outliers). The points are basically pulling the regression line and likely causing that massive variance, or at least factoring into the magnitude of it. Thus, overall, there is evidence of non constant variance since the spread of residuals appears to increase for larger fitted values.

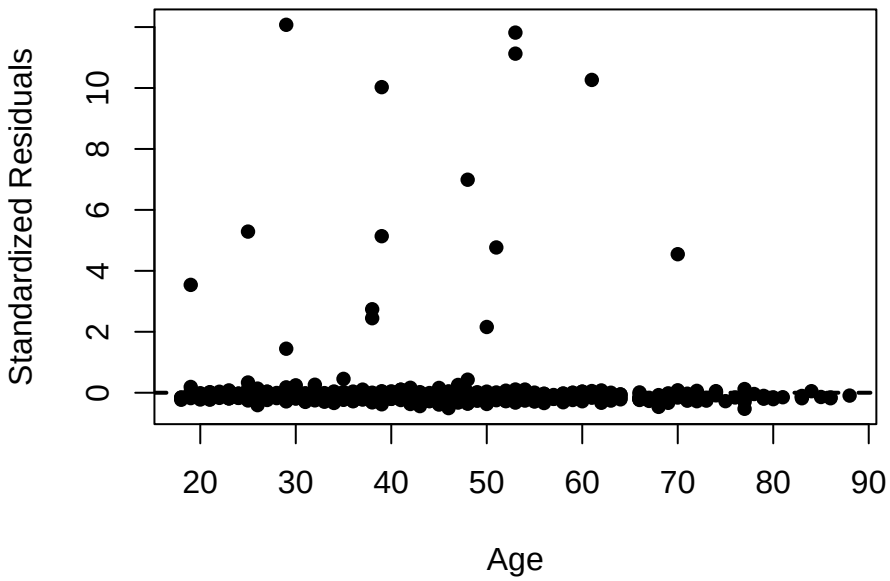
```r
library(faraway)
full_model <- lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc +
    Sibs, data = gss)
Y = gss$Income
X = model.matrix(full_model)
n = dim(X)[1]
p = dim(X)[2] - 1
residuals = full_model$residuals
plot(fitted(full_model), residuals, xlab = "Fitted Values", ylab = "Residuals",
    main = "Residuals vs Fitted")
abline(h = 0, col = "red")
```



Quantitative Covariates

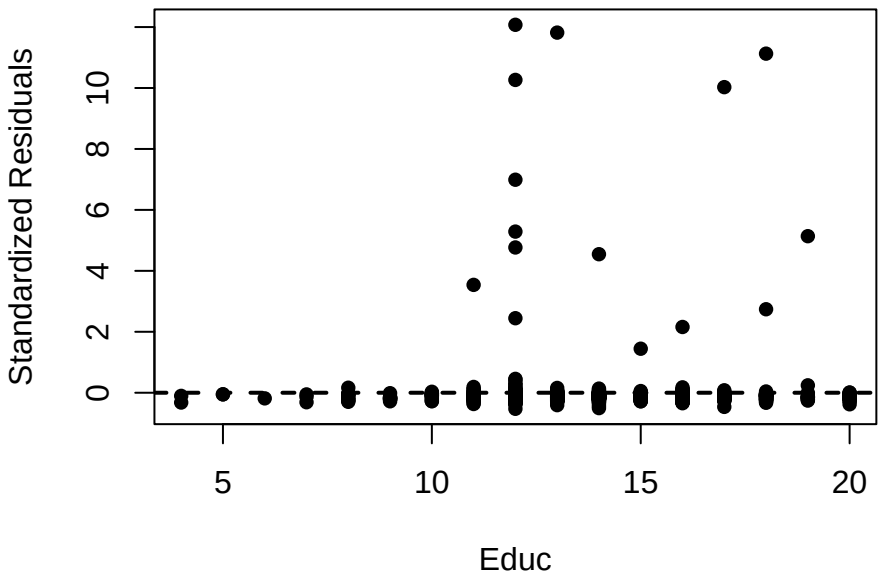
Responder's age

```r
plot(rstandard(full_model) ~ Age, data = gss, ylab = "Standardized Residuals",
    xlab = "Age", pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



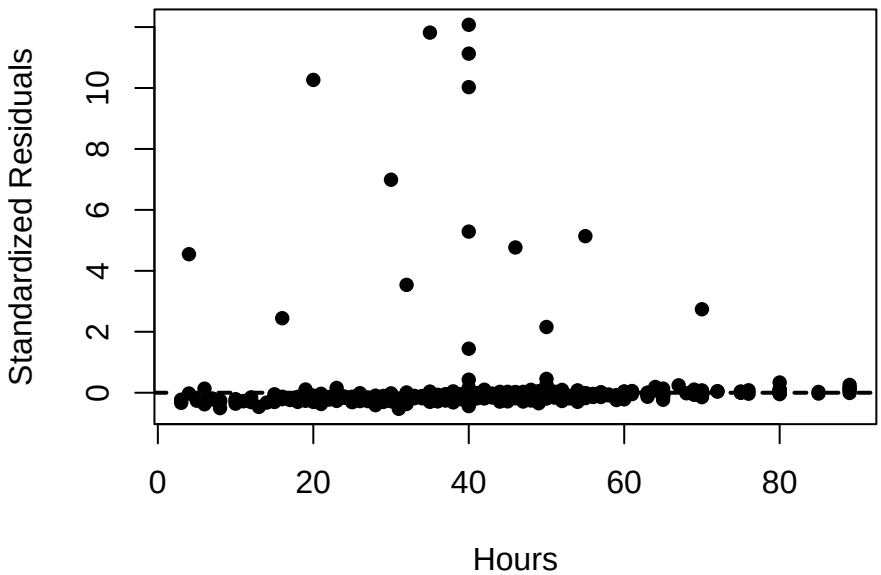
Total years of education

```
plot(rstandard(full_model) ~ Educ, data = gss, ylab = "Standardized Residuals",
    xlab = "Educ", pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



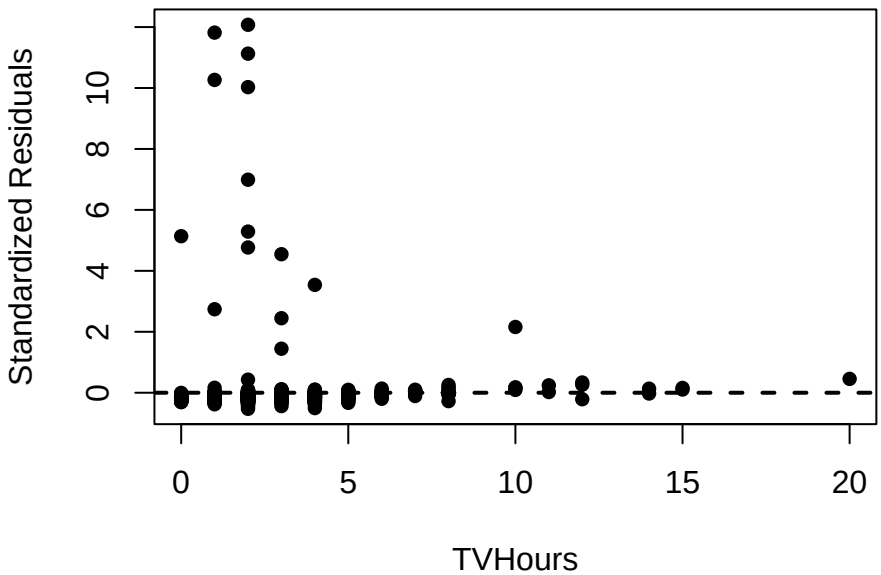
Weekly work hours

```
plot(rstandard(full_model) ~ Hours, data = gss, ylab = "Standardized Residuals",
    xlab = "Hours", pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



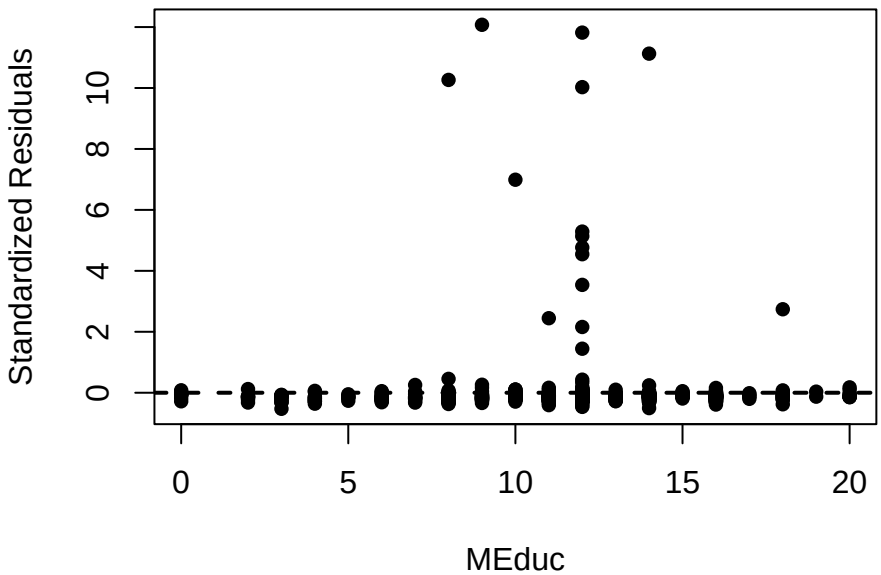
Hours spent watching TV daily

```
plot(rstandard(full_model) ~ TVHours, data = gss, ylab = "Standardized Residuals",
    xlab = "TVHours", pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



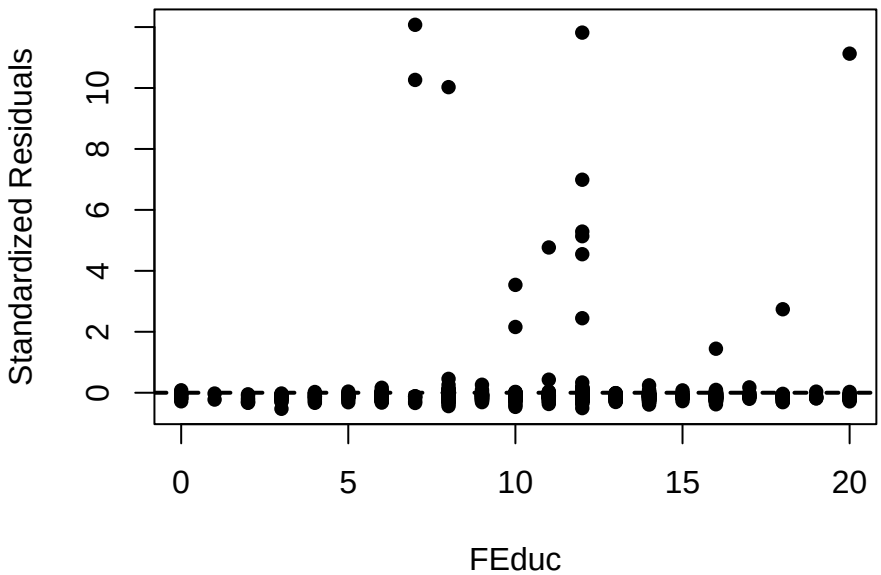
Mother's years of education

```
plot(rstandard(full_model) ~ MEduc, data = gss, ylab = "Standardized Residuals",
    xlab = "MEduc", pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



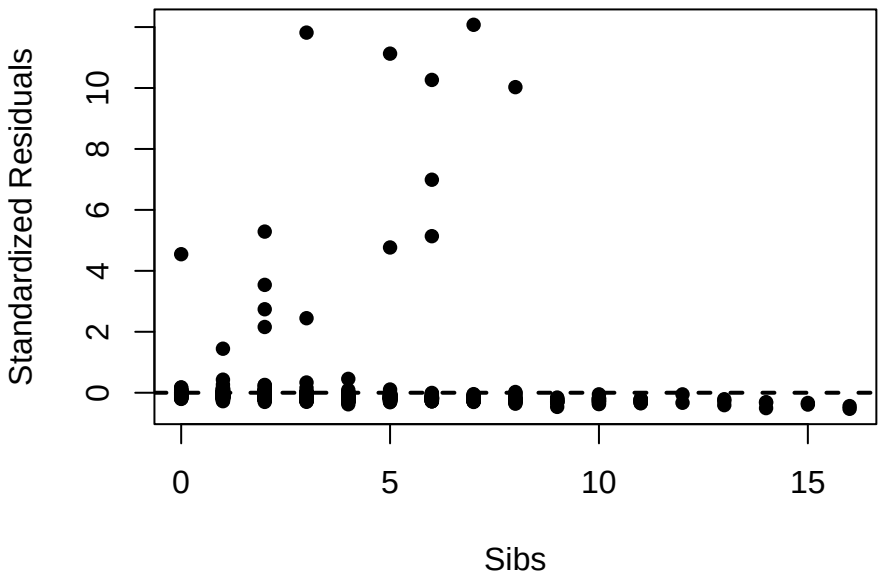
Father's years of education

```
plot(rstandard(full_model) ~ FEduc, data = gss, ylab = "Standardized Residuals",
    xlab = "FEduc", pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



Number of siblings

```
plot(rstandard(full_model) ~ Sibs, data = gss, ylab = "Standardized Residuals",
    xlab = "Sibs", pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```

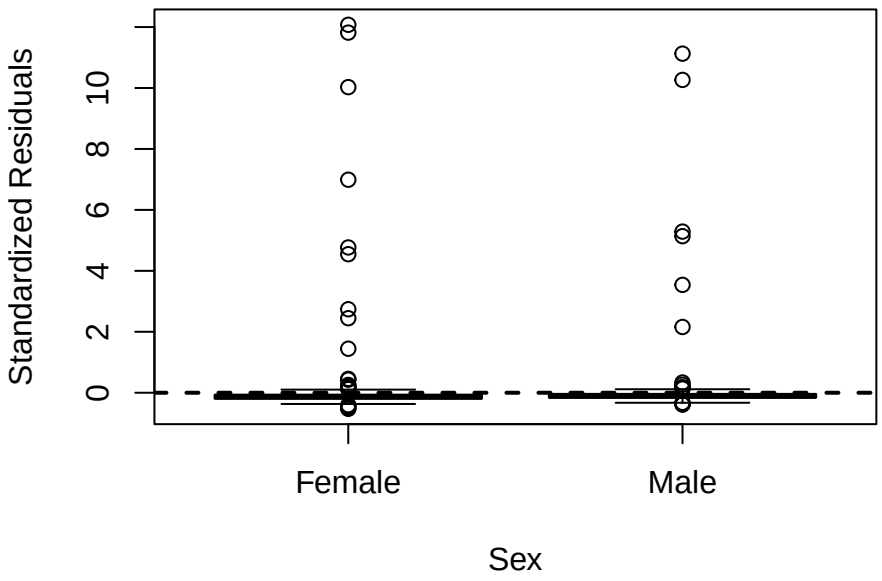


After inspecting the standardized residuals plotted against each quantitative covariate, I noticed the presence of extreme outliers but no systematic pattern of increasing or decreasing spread as a function of a quantitative covariate that would indicate the form of heteroscedasticity that appears when the variance of the random error terms is a function of some of the available covariates.

Qualitative Covariates

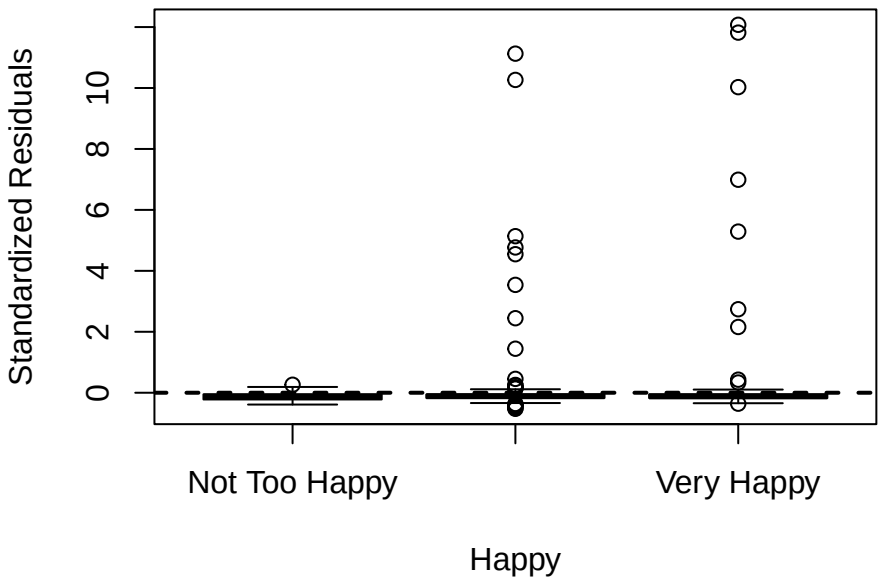
Responder's sex

```
boxplot(rstandard(full_model) ~ Sex, data = gss, ylab = "Standardized Residuals",
    xlab = "Sex")
abline(h = 0, lty = 2, lwd = 2)
```



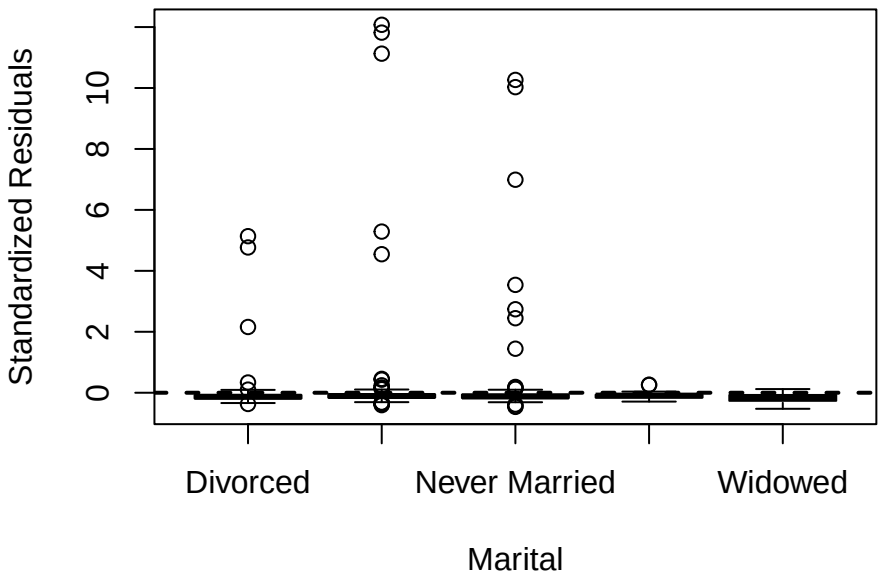
Happiness level

```
boxplot(rstandard(full_model) ~ Happy, data = gss, ylab = "Standardized Residuals",
    xlab = "Happy")
abline(h = 0, lty = 2, lwd = 2)
```



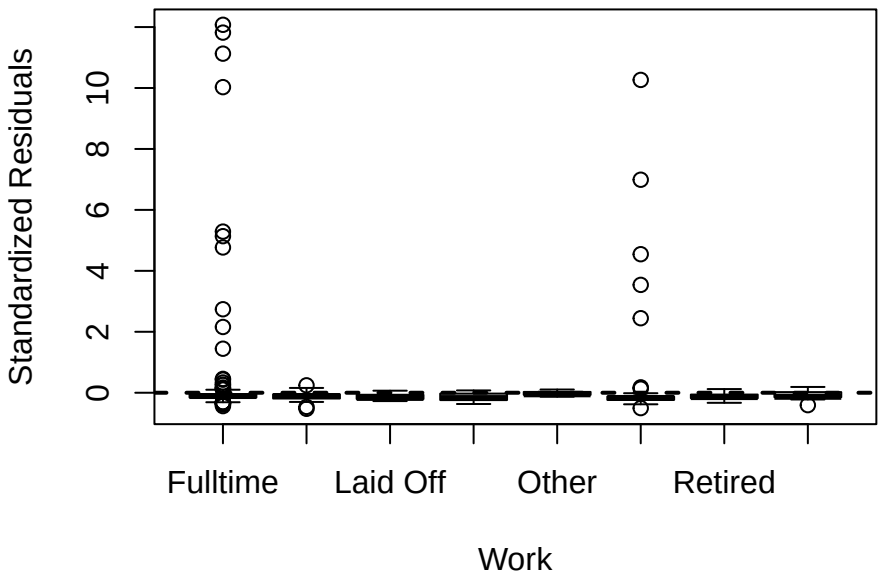
Marital status

```
boxplot(rstandard(full_model) ~ Marital, data = gss, ylab = "Standardized Residuals",
    xlab = "Marital")
abline(h = 0, lty = 2, lwd = 2)
```



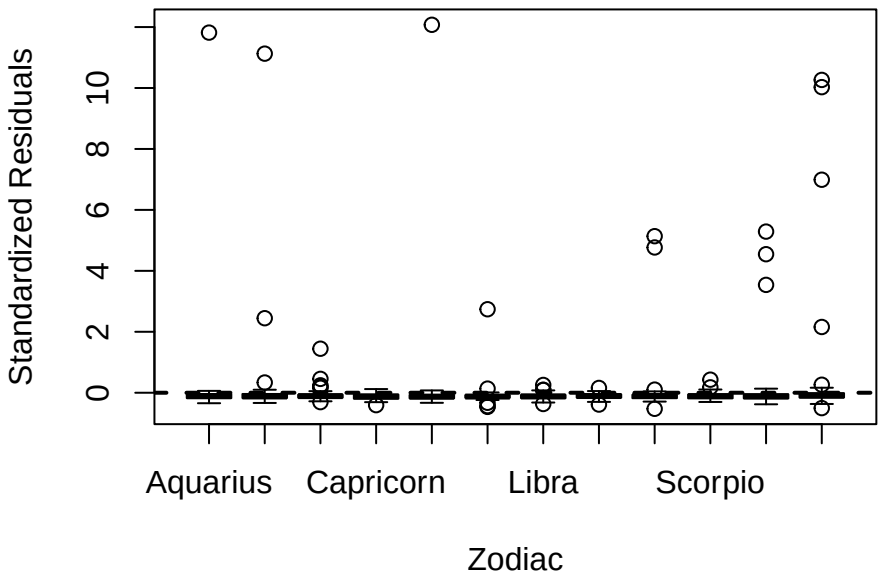
Work status

```
boxplot(rstandard(full_model) ~ Work, data = gss, ylab = "Standardized Residuals",
    xlab = "Work")
abline(h = 0, lty = 2, lwd = 2)
```



Zodiac sign

```
boxplot(rstandard(full_model) ~ Zodiac, data = gss, ylab = "Standardized Residuals",
    xlab = "Zodiac")
abline(h = 0, lty = 2, lwd = 2)
```



After inspecting the standardized residuals plotted against each qualititative covariate, I again noticed the presence of extreme outliers but no pattern of difference in spread for the boxplots across the levels of the qualitative covariate that would indicate the form of heteroscedasticity that appears when the variance of the random error terms differs across the different levels of some qualitative covariate.

So while we see visual indications for the violation of homoscedasticity in the residuals vs fitted plot, the pattern is less clear when the standardized residuals are plotted against individual quantitative and qualitative covariates.

ii) Perform suitable heteroscedasticity tests and state your conclusions.

We use the Breush-Pagan test for heteroscedasticity to test whether the variance of the error terms is a linear function of the covariates. The BP test statistic is 8.955868 and p-value is 0.2558401. Thus, we can not reject the null hypothesis that the full_model is homoscedastic since the p-value is larger than the alpha threshold of 0.05. This means we do not have sufficient evidence to reject the null hypothesis of homoskedasticity at $\alpha = 5\%$ and we do not detect a relationship of linear dependence of variance on the covariates.

```
# Breusch-Pagan Test for heteroscedasticity
Y = gss$Income
residuals = full_model$residuals
Yaux = residuals^2
aux = lm(Yaux ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs,
    data = gss)
```

```r
BP = n * summary(aux)$r.squared
print(BP)
```

```
## [1] 8.955868
```

```r
pvalue = pchisq(BP, p, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.2558401
```

```r
library(lmtest)
bptest(full_model)
```

```
##
##  studentized Breusch-Pagan test
##
## data:  full_model
## BP = 8.9559, df = 7, p-value = 0.2558
```

We can also use the Brown-Forsythe Test to test for heteroscedasticity-

Brown-Forsythe for Sex:

Grouping by sex, since we need to group by a categorical variable with levels (sex has 2 levels so it's valid). So, for sex the Brown-Forscythe test shows that there is np sufficient evidence that it is heteroscadastic since the p-value is 0.6143.

```r
gss$Sex <- as.factor(gss$Sex)
fit = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs,
    data = gss)
Ymed = aggregate(Income ~ Sex, data = gss, median)[gss$Sex, 2]
Yaux = abs(gss$Income - Ymed)
aux = lm(Yaux ~ Sex, data = gss)
anova(aux)
```

```
## Analysis of Variance Table
##
## Response: Yaux
##            Df     Sum Sq    Mean Sq F value Pr(>F)
## Sex         1 1.3923e+12 1.3923e+12  0.2542 0.6143
## Residuals 813 4.4533e+15 5.4776e+12
```

```r
library(car)
leveneTest(Income ~ Sex, data = gss)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.2542 0.6143
##       813
```

Brown-Forsythe for Work: Grouping by work, since we need to group by a categorical variable that has levels (Work has 2+). So, for work the Brown-Forscythe test shows that there is no sufficient evidence that it is heteroscadastic since the p-value is 0.6582.

```r
gss$Work <- as.factor(gss$Work)
fit = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs,
    data = gss)
Ymed = aggregate(Income ~ Work, data = gss, median)[gss$Work, 2]
Yaux = abs(gss$Income - Ymed)
aux = lm(Yaux ~ Work, data = gss)
anova(aux)
```

```
## Analysis of Variance Table
##
## Response: Yaux
##            Df     Sum Sq    Mean Sq F value Pr(>F)
## Work        7 2.7508e+13 3.9297e+12  0.7163 0.6582
## Residuals 807 4.4272e+15 5.4860e+12
```

```r
library(car)
leveneTest(Income ~ Work, data = gss)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   7  0.7163 0.6582
##       807
```

Brown-Forsythe for Marital: Grouping by Marital, since we need to group by a categorical variable that has levels (Marital has 2+). So, for work the Brown-Forscythe test shows that there is no sufficient evidence that it is heteroscadastic since the p-value is 0.7978.

```r
gss$Marital <- as.factor(gss$Marital)
fit = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs,
    data = gss)
Ymed = aggregate(Income ~ Marital, data = gss, median)[gss$Marital, 2]
Yaux = abs(gss$Income - Ymed)
aux = lm(Yaux ~ Marital, data = gss)
anova(aux)
```

```
## Analysis of Variance Table
```

```
##
## Response: Yaux
##            Df     Sum Sq    Mean Sq F value Pr(>F)
## Marital     4 9.1160e+12 2.2790e+12  0.4152 0.7978
## Residuals 810 4.4462e+15 5.4892e+12
```

```r
library(car)
leveneTest(Income ~ Marital, data = gss)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   4  0.4152 0.7978
##       810
```

Brown-Forsythe for Zodiac: Grouping by Zodiac, since we need to group by a categorical variable that has levels (Zodiac has 2+). So, for work the Brown-Forscythe test shows that there is no sufficient evidence that it is heteroscadastic since the p-value is 0.2985

```r
gss$Zodiac <- as.factor(gss$Zodiac)
fit = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs,
    data = gss)
Ymed = aggregate(Income ~ Zodiac, data = gss, median)[gss$Zodiac, 2]
Yaux = abs(gss$Income - Ymed)
aux = lm(Yaux ~ Zodiac, data = gss)
anova(aux)
```

```
## Analysis of Variance Table
##
## Response: Yaux
##            Df     Sum Sq    Mean Sq F value Pr(>F)
## Zodiac     11 7.0699e+13 6.4272e+12  1.1776 0.2985
## Residuals 803 4.3828e+15 5.4580e+12
```

```r
library(car)
leveneTest(Income ~ Zodiac, data = gss)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group  11  1.1776 0.2985
##       803
```

Brown-Forsythe for Happy: Grouping by Happy, since we need to group by a categorical variable that has levels (Happy has 2+). So, for work the Brown-Forscythe test shows that there is no sufficient evidence that it is heteroscadastic since the p-value is 0.2187

```r
gss$Happy <- as.factor(gss$Happy)
fit = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs,
    data = gss)
Ymed = aggregate(Income ~ Happy, data = gss, median)[gss$Happy, 2]
Yaux = abs(gss$Income - Ymed)
aux = lm(Yaux ~ Happy, data = gss)
anova(aux)
```

```
## Analysis of Variance Table
##
## Response: Yaux
##            Df     Sum Sq    Mean Sq F value Pr(>F)
## Happy       2 1.6645e+13 8.3226e+12  1.5231 0.2187
## Residuals 812 4.4370e+15 5.4642e+12
```

```r
library(car)
leveneTest(Income ~ Happy, data = gss)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   2  1.5231 0.2187
##       812
```

We can not reject the null hypothesis for every single variable here since all p-values are above 0.05 (in fact they are all above 0.2). The difference in spreads between our groups (e.g., Income variance between Men and Women) is likely due to random chance rather than a systematic difference for each of the variables. This technically means constant variance is achieved for these variables.

Step 3:

i) Calculate a suitable heteroscedasticity-consistent estimate for the covariance matrix of the regression coefficients. How does it compare against the corresponding ordinary least squares estimate?

ii) Calculate heteroscedasticity-consistent t-test statistics and 95 percent CIs for the regression coefficients. How do they compare against the results yielded by the ordinary least squares method?

```r
X = model.matrix(full_model)
n = dim(X)[1]
p = dim(X)[2]
beta = full_model$coefficients
residuals = full_model$residuals
vcovOLS = summary(full_model)$sigma^2 * summary(full_model)$cov.unscaled
print(vcovOLS)
```

```
##              (Intercept)           Age         Educ         Hours     TVHours
## (Intercept) 474220174987 -1907433705.2 -12064670132 -1347058849.0 -8426312775
## Age          -1907433705    38829681.0    -31202461     -432037.2   -15594169
## Educ        -12064670132   -31202461.2   1269225187   -20945952.6   200001770
## Hours        -1347058849     -432037.2    -20945953    35081614.4    15414790
## TVHours      -8426312775   -15594168.6    200001770    15414789.6  1805687552
## MEduc        -5079147870    29086917.4   -236010377      138585.3    28954807
## FEduc        -4109118474    37000088.9   -154559694     6151805.3    53859762
## Sibs         -9103007619    -7167222.2    167953096    14433306.1   -56165946
##                     MEduc       FEduc        Sibs
## (Intercept) -5079147869.7 -4109118474 -9103007619
## Age            29086917.4    37000089    -7167222
## Educ         -236010377.3  -154559694   167953096
## Hours            138585.3     6151805    14433306
## TVHours        28954807.1    53859762   -56165946
## MEduc        1069222108.8  -493576358    79738958
## FEduc        -493576358.5   825093313   159284579
## Sibs           79738957.8   159284579  1121549496
```

The covariance matrix of the least squares estimator is $\begin{pmatrix} 474220174987 & -1907433705.2 & -12064670132 & -134705 \\ -1907433705 & 38829681.0 & -31202461 & -4320 \\ -12064670132 & -31202461.2 & 1269225187 & -20945 \\ -1347058849 & -432037.2 & -20945953 & 35081 \\ -8426312775 & -15594168.6 & 200001770 & 15414 \\ -5079147870 & 29086917.4 & -236010377 & 13858 \\ -4109118474 & 37000088.9 & -154559694 & 61518 \\ -9103007619 & -7167222.2 & 167953096 & 144333 \end{pmatrix}$

```r
confint(full_model)
```

```
##                    2.5 %      97.5 %
## (Intercept) -1139787.136 1563672.818
## Age            -9282.145   15180.975
## Educ          -49482.462   90379.388
## Hours         -20949.732    2302.771
## TVHours      -148773.710   18047.486
## MEduc         -67900.262   60469.842
## FEduc         -46177.946   66588.907
## Sibs           -7087.921  124385.860
```

The 95% observed CI for the regression coefficient $\hat{\beta}_0$, the intercept, is [-1139787.136, 1563672.818]. The 95% observed CI for the regression coefficient$\hat{\beta}_{Age}$, is [-9282.145, 15180.975]. The 95% observed CI for the regression coefficient$\hat{\beta}_{Educ}$, is [-49482.462, 90379.388]. The 95% observed CI for the regression coefficient$\hat{\beta}_{Hours}$, is [-20949.732, 2302.771]. The 95% observed CI for the regression coefficient$\hat{\beta}_{TVHours}$, is [-148773.710, 18047.486]. The 95% observed CI for the regression coefficient$\hat{\beta}_{MEduc}$, is [-67900.262, 60469.842].The 95% observed CI for the regression coefficient$\hat{\beta}_{FEduc}$, is [-46177.946, 66588.907]. The 95% observed CI for the regression coefficient$\hat{\beta}_{Sibs}$, is [-7087.921, 124385.860].

```r
summary(full_model)
```

```
##
## Call:
## lm(formula = Income ~ Age + Educ + Hours + TVHours + MEduc +
##     FEduc + Sibs, data = gss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1211280  -410167  -271382  -129856 28149890
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   211943     688636   0.308   0.7583
## Age             2949       6231   0.473   0.6361
## Educ           20448      35626   0.574   0.5661
## Hours          -9324       5923  -1.574   0.1159
## TVHours       -65363      42493  -1.538   0.1244
## MEduc          -3715      32699  -0.114   0.9096
## FEduc          10206      28724   0.355   0.7225
## Sibs           58649      33490   1.751   0.0803 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2340000 on 807 degrees of freedom
## Multiple R-squared:  0.0101, Adjusted R-squared:  0.001516
## F-statistic: 1.177 on 7 and 807 DF,  p-value: 0.3136
```

The observed t test statistic of statistical significance for the intercept regression coefficient is 0.308 with p-value 0.7583. The observed t test statistic of statistical significance for the Age regression coefficient is 0.473 with p-value 0.6361. The observed t test statistic of statistical significance for the Educ regression coefficient is 0.574 with p-value 0.5661. The observed t test statistic of statistical significance for the Hours regression coefficient is -1.574 with p-value 0.1159. The observed t test statistic of statistical significance for the TVHours regression coefficient is -1.538 with p-value 0.1244. The observed t test statistic of statistical significance for the MEduc regression coefficient is -0.114 with p-value 0.9096. The observed t test statistic of statistical significance for the FEduc regression coefficient is 0.355 with p-value 0.7225. The observed t test statistic of statistical significance for the Sibs regression coefficient is 1.751 with p-value 0.0803. Thus, none of the regression coefficients are statistically significant at the 0.05 level.

i) Calculate a suitable heteroscedasticity-consistent estimate for the covariance matrix of the regression coefficients. How does it compare against the corresponding ordinary least squares estimate?

```r
Sigma = diag(residuals^2/(1 - hatvalues(full_model))^2)
vcovHC3 = summary(full_model)$cov.unscaled %*% crossprod(X, Sigma) %*%
    X %*% summary(full_model)$cov.unscaled
print(vcovHC3)
```

```
##              (Intercept)         Age         Educ       Hours     TVHours
## (Intercept) 647905975588 -2119149267 -19837922241 -1357649949 -9486014566
## Age          -2119149267    48814436    -24590704   -17494011   -50260968
## Educ        -19837922241   -24590704   1162078662    31381166   143728482
## Hours        -1357649949   -17494011     31381166    27278306   103955035
## TVHours      -9486014566   -50260968    143728482   103955035  1240245743
## MEduc         3595276136   -34217774    -53411133     9117008    30651678
## FEduc       -17028391209   162045871     94715693    -7441799   103186441
## Sibs        -11637769372   -37033183    888805838    -9925909  -486253897
##                   MEduc        FEduc         Sibs
## (Intercept) 3595276136 -17028391209 -11637769372
## Age          -34217774    162045871    -37033183
## Educ         -53411133     94715693    888805838
## Hours          9117008     -7441799     -9925909
## TVHours       30651678    103186441   -486253897
## MEduc        245096719   -386380314   -132459705
## FEduc       -386380314   1151032066    101992572
## Sibs        -132459705    101992572   1425697587
```

The heteroscedasticity-consistent estimate for the covariance matrix of the regression coefficients is

$$\begin{pmatrix} 647905975588 & -2119149267 & -19837922241 & -1357649949 & -9486014566 & 3595276136 & -17028391209 & - \\ -2119149267 & 48814436 & -24590704 & -17494011 & -50260968 & -34217774 & 162045871 & \\ -19837922241 & -24590704 & 1162078662 & 31381166 & 143728482 & -53411133 & 94715693 & \\ -1357649949 & -17494011 & 31381166 & 27278306 & 103955035 & 9117008 & -7441799 & \\ -9486014566 & -50260968 & 143728482 & 103955035 & 1240245743 & 30651678 & 103186441 & - \\ 3595276136 & -34217774 & -53411133 & 9117008 & 30651678 & 245096719 & -386380314 & - \\ -17028391209 & 162045871 & 94715693 & -7441799 & 103186441 & -386380314 & 1151032066 & \\ -11637769372 & -37033183 & 888805838 & -9925909 & -486253897 & -132459705 & 101992572 & \end{pmatrix}$$

The variance of the intercept increases from 474220174987 to 647905975588. The variance of the Age coefficient increases from 38829681.0 to 48814436 The variance of the Educ coefficient decreases from 1269225187 to 1162078662. The variance of the Hours coefficient decreases from 35081614.4 to 27278306. The variance of the TVHours coefficient decreases from 1805687552 to 1240245743. The variance of the MEduc coefficient decreases from 1069222108.8 to 245096719. The variance of the FEduc coefficient increases from 825093313 to 1151032066. The variance of the Sibs coefficient increases from 1121549496 to 1425697587.

So, the heteroscedasticity-consistent estimator overall produces broadly similar estimated variances and standard errors than the ordinary least squares estimator. The heteroscedasticity-consistent

estimate for the covariance matrix of the least squares estimator is similar compared to the one obtained through the least squares method due to the lack of evidence for heteroscedasticity in the random error terms.

ii) Calculate heteroscedasticity-consistent t-test statistics and 95 percent CIs for the regression coefficients. How do they compare against the results yielded by the ordinary least squares method?

Heteroscedasticity-consistent t-test statistics

```r
library(lmtest)
coeftest(full_model, vcov. = vcovHC3)
```

```
##
## t test of coefficients:
##
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 211942.8   804926.1  0.2633  0.79238
## Age           2949.4     6986.7  0.4221  0.67303
## Educ         20448.5    34089.3  0.5999  0.54877
## Hours        -9323.5     5222.9 -1.7851  0.07462 .
## TVHours     -65363.1    35217.1 -1.8560  0.06382 .
## MEduc        -3715.2    15655.6 -0.2373  0.81248
## FEduc        10205.5    33926.9  0.3008  0.76364
## Sibs         58649.0    37758.4  1.5533  0.12075
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

For the intercept coefficient, the heteroscedasticity-consistent t test statistic is 0.2633, and corresponding p-value is 0.79238. Compared to the t-statistic and p-value yielded by the ordinary least squares method, 0.308 and 0.7583, respectively, the heteroscedasticity-consistent t-test statistic is slightly less extreme with a slightly larger p-value.

For the Age coefficient, the heteroscedasticity-consistent t test statistic is 0.4221, and corresponding p-value is 0.67303. Compared to the t-statistic and p-value yielded by the ordinary least squares method, 0.473 with p-value 0.6361, respectively, the heteroscedasticity-consistent t-test statistic is slightly less extreme with a slightly larger p-value.

For the Educ coefficient, the heteroscedasticity-consistent t test statistic is 0.5999, and corresponding p-value is 0.54877 Compared to the t-statistic and p-valye yielded by the ordinary least squares method, 0.574 with p-value 0.5661, respectively, the heteroscedasticity-consistent t-test statistic is slightly more extreme with a slightly smaller p-value.

For the Hours coefficient, the heteroscedasticity-consistent t test statistic is -1.7851, and corresponding p-value is 0.07462 Compared to the t-statistic and p-value yielded by the ordinary least squares method, -1.574 with p-value 0.1159, respectively, the heteroscedasticity-consistent t-test statistic is slightly more extreme with a slightly smaller p-value.

For the TVHours coefficient, the heteroscedasticity-consistent t test statistic is -1.8560, and corresponding p-value is 0.06382 Compared to the t-statistic and p-value yielded by the ordinary least squares method, -1.538 with p-value 0.1244, respectively, the heteroscedasticity-consistent t-test statistic is more extreme with a slightly smaller p-value.

For the MEduc coefficient, the heteroscedasticity-consistent t test statistic is -0.2373, and corresponding p-value is 0.81248. Compared to the t-statistic and p-value yielded by the ordinary least squares method, -0.114 with p-value 0.9096, respectively, the heteroscedasticity-consistent t-test statistic is slightly more extreme with a slightly smaller p-value.

For the FEduc coefficient, the heteroscedasticity-consistent t test statistic is 0.3008, and corresponding p-value is 0.76364 Compared to the t-statistic and p-value yielded by the ordinary least squares method, 0.355 with p-value 0.7225, respectively, the heteroscedasticity-consistent t-test statistic is slightly less extreme with a slightly larger p-value.

For the Sibs coefficient, the heteroscedasticity-consistent t test statistic is 1.5533, and corresponding p-value is 0.12075 Compared to the t-statistic and p-value yielded by the ordinary least squares method, 1.751 with p-value 0.0803, respectively, the heteroscedasticity-consistent t-test statistic is less extreme with a larger p-value.

Both methods do not produce statistically significant p-values for all coefficients.

95 percent CIs

```r
alpha = 0.05
CI = cbind(beta - qt(1 - alpha/2, n - p) * sqrt(diag(vcovHC3)), beta +
    qt(1 - alpha/2, n - p) * sqrt(diag(vcovHC3)))
colnames(CI) = colnames(confint(full_model))
print(CI)
```

```
##                   2.5 %       97.5 %
## (Intercept) -1368052.93 1791938.6124
## Age           -10764.90   16663.7286
## Educ          -46465.65   87362.5712
## Hours         -19575.48     928.5205
## TVHours      -134491.08    3764.8580
## MEduc         -34445.64   27015.2230
## FEduc         -56389.83   76800.7903
## Sibs          -15467.31  132765.2539
```

The heteroscedasticity-consistent 95% CIs for the intercept coefficient is [-1368052.93, 1791938.6124]. Compared to the results yielded by the ordinary least squares method [-1139787.136, 1563672.818], the heteroscedasticity-consistency interval is wider.

The heteroscedasticity-consistent 95% CIs for the Age coefficient is [-10764.90, 16663.7286]. Compared to the results yielded by the ordinary least squares method [-9282.145, 15180.975], the heteroscedasticity-consistency interval is wider.

The heteroscedasticity-consistent 95% CIs for the Educ coefficient is [-46465.65, 87362.5712]. Compared to the results yielded by the ordinary least squares method [-49482.462, 90379.388], the heteroscedasticity-consistency interval is similar

The heteroscedasticity-consistent 95% CIs for the Hours coefficient is [-19575.48, 928.5205]. Compared to the results yielded by the ordinary least squares method [-20949.732, 2302.771], the heteroscedasticity-consistency interval is slightly narrower.

The heteroscedasticity-consistent 95% CIs for the TVHours coefficient is [-134491.08, 3764.8580]. Compared to the results yielded by the ordinary least squares method [-148773.710, 18047.486], the heteroscedasticity-consistency interval is narrower

The heteroscedasticity-consistent 95% CIs for the MEduc coefficient is [-34445.64, 27015.2230]. Compared to the results yielded by the ordinary least squares method [-67900.262, 60469.842], the heteroscedasticity-consistency interval is much more narrow.

The heteroscedasticity-consistent 95% CIs for the FEduc coefficient is [-56389.83, 76800.7903]. Compared to the results yielded by the ordinary least squares method [-46177.946, 66588.907], the heteroscedasticity-consistency interval is slightly wider

The heteroscedasticity-consistent 95% CIs for the Sibs coefficient is [-15467.31, 132765.2539]. Compared to the results yielded by the ordinary least squares method [-7087.921, 124385.860], the heteroscedasticity-consistency interval is much wider.

The heteroscedasticity-consistent 95% CIs for Hours, TVHours MEduc are narrower than the 95% CIs by the ordinary least squares method, so the OLS may have overestimated variability there, but overall the heteroscedasticity-consistent 95% CIs are mostly wider than the 95% CIs by the ordinary least squares method. Still all methods contain 0 so the conclusions are the same at the 5% level and none of the regression coefficients are statistically significant.

Patterns

The variables Hours, TVHours, and especially MEduc, consistently exhibit the most changes from OLS to HC, where their test statistics become more extreme, the variance of their coefficients decrease, and their confidence intervals become substantially narrower.

Step 4:

i) Apply the iteratively reweighted least squares method to specify optimal weights for the variances of the random error terms as a function of the quantitative covariate which has been specified for your team.

ii) Write down and interpret your estimated equation for the residual variance.

iii) Use the optimal weights to apply the weighted least squares method with all available quantitative covariates.

iv) Calculate t test statistics and 95 percent CIs for the regression coefficients.

v) How do they compare against the results yielded by the ordinary least squares method?

---

i) Apply the iteratively reweighted least squares method to specify optimal weights for the variances of the random error terms as a function of the quantitative covariate which has been specified for your team.

Weighted Least Squares

```r
w = gss$Income^(-2)  # model is income based
WLS = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs,
    data = gss, weights = w)
summary(WLS$fitted.values)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   -4181    1974    4453    4801    7278   20006
```

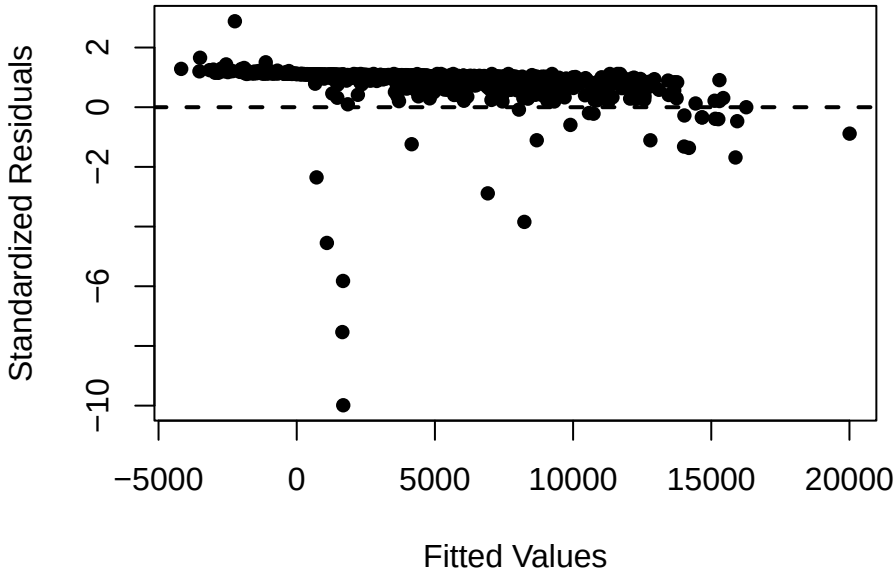
```r
summary(summary(WLS)$residuals)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -3.1181  0.7923  0.9089  0.7981  0.9640  2.1192
```

```r
summary(WLS)
```

```
##
## Call:
## lm(formula = Income ~ Age + Educ + Hours + TVHours + MEduc +
##     FEduc + Sibs, data = gss, weights = w)
##
## Weighted Residuals:
##     Min      1Q  Median      3Q     Max
## -3.1181  0.7923  0.9089  0.9640  2.1192
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -5603.43    2933.63  -1.910  0.05648 .
## Age           229.07      30.90   7.414 3.11e-13 ***
## Educ           16.87      60.15   0.281  0.77913
## Hours         -69.61      25.39  -2.742  0.00624 **
## TVHours      -100.96      98.27  -1.027  0.30454
## MEduc        -336.67     124.25  -2.710  0.00688 **
## FEduc         471.23      70.81   6.655 5.22e-11 ***
## Sibs          607.76     117.46   5.174 2.89e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8978 on 807 degrees of freedom
## Multiple R-squared:  0.1221, Adjusted R-squared:  0.1145
## F-statistic: 16.04 on 7 and 807 DF,  p-value: < 2.2e-16
```

```r
plot(WLS$fitted.values, rstandard(WLS), xlab = "Fitted Values", ylab = "Standardized Residuals"
    pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



We finally have gotten statistically significant covariates and respectively significant p-values through the WLS model. Age has a statistically significant positive effect on income under WLS, having coefficient 229.07 and p-value 3.11e-13. Hours has a statistically significant negative effect on income under WLS, having coefficient -69.61 and p-value 0.00624. MEduc has a statistically significiant negative effect on income under WLS, having coefficient -336.67 and p-value 0.00688. FEduc has a statistically significant positive effect on income under WLS, having coefficient 471.23 and p-value 5.22e-11. And lastly, Sibs has a statistically significant positive effect on income under WLS with coefficient 607.76 and p-value 2.89e-07. Up until this point, we had no variables show statistical significance and because WLS reweights to account for possible heteroscedasticity, the fact that several covariates became significant under WLS may indicate that non-constant variance in the OLS model masked these relationships.

If high-income individuals in the GSS have massive variance in their reported income (some making 100k, some 500k with the same education level), OLS treats that noise as if it applies to everyone. However, WLS clears out that noise and puts less emphasis on the data points with extremely high variance (the outliers/ leverage points) and gives more emphasis to the stable/ "good" data points. The WLS model is: 1. Isolating the noise 2. Decreases standard error (through giving emphasis to stable data points) 3. P-values then drop below 0.05 due to the reasons above and we finally see which covariates are truly significant

WLS doesn't create significance rather it just fixes a violation of Gauss-Markov assumptions that was hiding the true relationship.

Iteratively Reweighted Least Squares Method as a function of the Age Covariate

```r
w = rep(1, n)
err = Inf
tol = 1e-10
niter = 0
maxiter = 1000

while (err > tol & niter < maxiter) {
    aux = lm(log(residuals^2) ~ log(Age), data = gss)

    err = sum(abs(w - exp(-aux$fitted.values)))/sum(w)
    w = exp(-aux$fitted.values)  # model is now aged based weights
    WLS = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc +
        Sibs, data = gss, weights = w)

    residuals = WLS$residuals
    niter = niter + 1
}
summary(aux)
```

```
##
## Call:
## lm(formula = log(residuals^2) ~ log(Age), data = gss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.1868  -0.7739   0.1970   0.9735   9.6953
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  22.2574     0.7599  29.289  < 2e-16 ***
## log(Age)      0.6978     0.2044   3.414 0.000671 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.977 on 813 degrees of freedom
## Multiple R-squared:  0.01414,    Adjusted R-squared:  0.01292
## F-statistic: 11.66 on 1 and 813 DF,  p-value: 0.0006708
```

```r
summary(WLS)
```

```
##
## Call:
## lm(formula = Income ~ Age + Educ + Hours + TVHours + MEduc +
```

```
##     FEduc + Sibs, data = gss, weights = w)
##
## Weighted Residuals:
##     Min      1Q  Median      3Q     Max
##  -4.477  -1.541  -1.052  -0.583 127.440
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   263813     661484   0.399   0.6901
## Age             2073       6360   0.326   0.7446
## Educ           19811      35432   0.559   0.5762
## Hours          -7530       5832  -1.291   0.1970
## TVHours       -56355      40846  -1.380   0.1681
## MEduc          -2459      32197  -0.076   0.9391
## FEduc           -694      28387  -0.024   0.9805
## Sibs           59813      33614   1.779   0.0755 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.297 on 807 degrees of freedom
## Multiple R-squared:  0.008883,   Adjusted R-squared:  0.0002864
## F-statistic: 1.033 on 7 and 807 DF,  p-value: 0.406
```

The summary of the aux model confirms that Age is statistically associated with the variance of residuals in the GSS data since it produces a p-value of 0.00067. The slope is also positive, 0.6978 which implies that variance increases with age, but at a relatively low rate, suggesting that income variability grows as people get older.

After applying the IRLS method using age-based weights, the resulting model yields no statistically significant covariates at the 0.05 level. The R^2 adj is at 0.0002864 which basically implies that the model explains none of the variation in income.

This likely happened because the IRLS procedure models the variance as a function of Age which may not effectively capture the heteroscedasticity of the data. Thus we can see that the standard errors increased for example Age having a now having an SE of 6360 rather than 2073 as before. These inflated errors and loss of significance is consistent with the very low adjusted R-squared value.

Thus it is very important to correctly specify the variance when applying WLS as the wrong specification can worsen the model.

ii) Write down and interpret your estimated equation for the residual variance.

Estimated equation for the residual variance: $\log(\hat{\sigma_i^2}) = \hat{\alpha_0} + 0.6978 \log(\text{Age})$ Exponentiate: $\sigma_i^2 = \exp^{\hat{\alpha_0}} * \text{Age}^{0.6978}$

```r
# From the aux in part i
intercept <- exp(22.2574)
```

```r
multiplier <- gss$Age^(0.6978)

residual_variance = intercept * multiplier

new_weight = 1/(residual_variance)
```

The coefficient is positive which means that the variance of the error term increases as Age increases. Each age data point (within the 815) is put to the power of 0.6978, meaning that the power exponent of the income residuals are set to a magnified by 0.7 (it is the magnitude multiplier).

iii) Use the optimal weights to apply the weighted least squares method with all available quantitative covariates.

```r
WLS = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs,
    data = gss, weights = new_weight)
summary(WLS$fitted.values)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -715463  199270  314758  324681  434852 1262554
```

```r
summary(summary(WLS)$residuals)
```

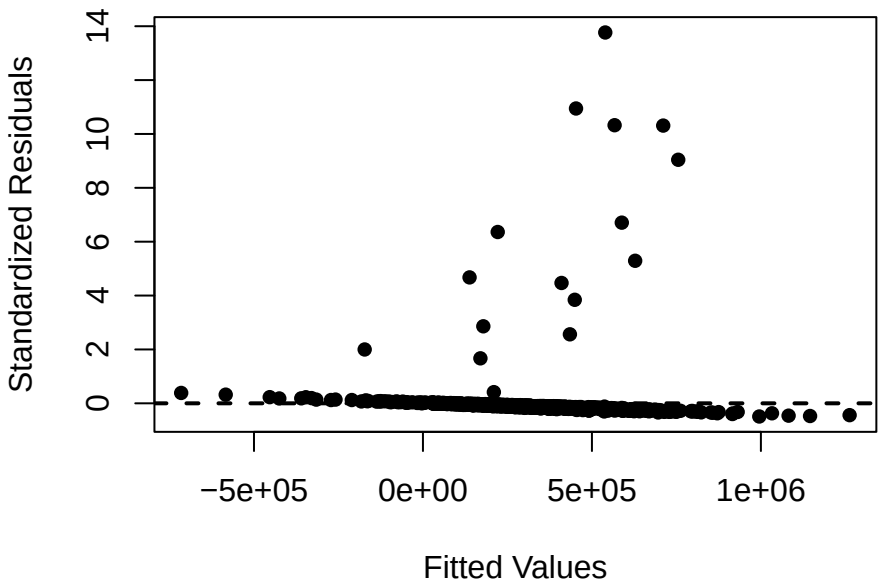
```
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
##  -4.477517  -1.541252  -1.052396   0.000172  -0.583248 127.442519
```

```r
summary(WLS)
```

```
##
## Call:
## lm(formula = Income ~ Age + Educ + Hours + TVHours + MEduc +
##     FEduc + Sibs, data = gss, weights = new_weight)
##
## Weighted Residuals:
##     Min      1Q  Median      3Q     Max
##  -4.478  -1.541  -1.052  -0.583 127.443
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 263811.3   661484.2   0.399   0.6901
## Age           2072.6     6359.8   0.326   0.7446
## Educ         19811.2    35431.6   0.559   0.5762
## Hours        -7529.7     5831.9  -1.291   0.1970
## TVHours     -56354.8    40846.3  -1.380   0.1681
```

```
## MEduc        -2459.0    32196.6  -0.076   0.9391
## FEduc         -693.8    28386.7  -0.024   0.9805
## Sibs         59813.0    33613.9   1.779   0.0755 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.297 on 807 degrees of freedom
## Multiple R-squared:  0.008883,   Adjusted R-squared:  0.0002864
## F-statistic: 1.033 on 7 and 807 DF,  p-value: 0.406
```

```r
plot(WLS$fitted.values, rstandard(WLS), xlab = "Fitted Values", ylab = "Standardized Residuals"
    pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



The spread on the Age based IRLS weighted residuals plot is larger than that from the income based IRLS weighted residuals plot. We can also see a new pattern where there is an increasing spread, and larger residuals at higher fitted values. So, after applying the iteratively reweighted least squares method to specify optimal weights as a function of the age covariate, we can see that variance increased and performance increased.

iv) Calculate t test statistics and 95 percent CIs for the regression coefficients.

T test statistics for the regression coefficients

```
coef_table <- summary(WLS)$coefficients
variable_t_stats <- coef_table[, "t value"]
variable_t_stats
```

```
## (Intercept)         Age        Educ       Hours     TVHours       MEduc
##  0.39881724  0.32588788  0.55913996 -1.29112793 -1.37967842 -0.07637364
##       FEduc        Sibs
## -0.02443995  1.77941445
```

95 percent CIs for the regression coefficients

```
WLS_ci <- confint(WLS, level = 0.95)
WLS_ci
```

```
##                    2.5 %     97.5 %
## (Intercept) -1034621.314 1562243.93
## Age           -10411.075   14556.22
## Educ          -49737.753   89360.20
## Hours         -18977.097    3917.73
## TVHours      -136532.284   23822.77
## MEduc         -65657.914   60739.97
## FEduc         -56414.177   55026.64
## Sibs           -6167.926  125793.93
```

v) How do they compare against the results yielded by the ordinary least squares method?

They all responded fairly similar to the OLS model in terms of T-Test Statistics with the soul exception of Female Education which changed signs. Female education went from 0.3 to -0.02. But since it is so low (close to 0), it essentially means that it is a non-significant covariate that we are still utilizing in the model. The other biggest change in the t-test statistic was TVHours and Hours, since they changed by 0.5 (becoming more negative). Yet, through all of our quantitiative covariates, none of them changed to be significant in any way. All other test statistics changed by no more than 0.15 beside those three covariates, demonstrating that the new weight based off the age did not "move the needle" for our potential inconsistencies.

That is also visible in the confidence intervals we constructed above. All of them contain zero, which means we can NOT reject the null hypothesis because we do not have sufficient evidence to prove it wrong otherwise. The results simply are not statistically significant, at least at the 5% level. It matches the insignificance that we identified in the OLS model, meaning that the weights based

on Age did not really alter our results to make anything significant. The entire model supplied a p-value of 0.4, which is even larger than the initial full models 0.15 p-value. Meaning our data has gotten less significant.

Step 5:

i) Ignore any heteroscedasticity issues from now on. Apply the jackknife and a suitable bootstrap method to estimate the standard errors of your regression coefficients.

ii) How do your estimates compare against the estimates obtained through the ordinary least squares method?

iii) Use the bootstrap estimates to construct different types of 95 percent CIs for the regression coefficients.

iv) How do the results of these methods compare against each other?

---

i) Ignore any heteroscedasticity issues from now on. Apply the jackknife and a suitable bootstrap method to estimate the standard errors of your regression coefficients.
Jackknife Method
“‘ r # re-establishing the full model full_model = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs, data = gss)
betajack = matrix(0, n, p)
for (i in 1:n) { jack = lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs, data = gss, subset = setdiff(1:n, i)) betajack[i,] = jack$coefficients }
betabar = colMeans(betajack) print(betabar) “‘
```
## [1] 211951.619   2949.435  20448.163  -9323.541 -65364.646  -3715.195
10205.343 ## [8]  58649.742
r SE = apply(betajack, 2, sd) * sqrt((n - 1)^2/n) print(SE)
## [1] 804432.058   6982.445  34068.354   5219.658  35195.483  15645.958
33906.043 ## [8]  37735.233
```
Standard Errors for the Jackknife Method: Intercept: 804432.058 Age: 6982.445 Educ: 34068.354 Hours: 5219.658 TVHours: 35195.483 MEduc: 15645.958 FEduc: 33906.043 Sibs: 37735.233
Choosing the nonparametric boostrap method makes the most sense rather than semi-parametric and parametric for a couple reasons: 1. It is the safest choice when the data is not normal or has outliers and as we have seen in earlier parts we have a lot of outliers and leverage points 2. Our plots showed outliers and potentianl non-constant variance, so this generates the most robust starting point for us to go off of.
The semi-parametric is not a bad choice either though since, at this point, we are fairly certain that the full_model is fairly correct, since nothing has come back saying that the covariates are significant or that the model itself is significant. But it does not acknowledge the outliers as well as the nonparametric does, and since we are currently ignoring potential heteroscadastity for the sake of analysis using the nonparametric test is the most useful. We also should not use the parametric test since we are not confident that our data is normally distributed given the amount of outliers and leverage points that we have riddled throughout our data.

Nonparametric Boostrap Method

```
r nboot = 10000 betaboot = matrix(0, nboot, p) for (k in 1:nboot) { boot =
lm(Income ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs, data = gss,
sample(n, replace = TRUE)) betaboot[k, ] = boot$coefficients } betabar =
colMeans(betaboot) print(betabar)
## [1] 217169.145   2867.527  20403.820  -9335.836 -65934.340  -3817.674
10151.481 ## [8]  59538.328
r SE = apply(betaboot, 2, sd) print(SE)
## [1] 806750.840   6981.644  34065.248   5138.233  35747.731  16089.728
34082.783 ## [8]  37859.148
```

Standard Errors for the Nonparametric Bootstrap Method: Intercept: 806702.926 Age: 6981.679 Educ: 34064.401 Hours: 5138.218 TVHours: 35746.581 MEduc: 16093.799 FEduc: 34084.656 Sibs: 37862.515

ii) How do your estimates compare against the estimates obtained through the ordinary least squares method?

```
r coef_full_model <- summary(full_model)$coefficients coef_std <-
coef_full_model[, "Std. Error"] coef_std
## (Intercept)         Age        Educ       Hours     TVHours       MEduc ##
688636.461    6231.347   35626.187    5922.973   42493.382   32698.962 ##
FEduc        Sibs ##   28724.438   33489.543
```

Standard Errors for the OLS Method proived by coef_std

Jackknife V.S. OLS Standard Error estimates

The Jackknife has a higher standard error for the intercept estimate than the OLS method, 804432.058 vs 661484.217. It is almost 115000 dollars larger than OLS's. Age is also slightly under evaluated in the OLS model, which means the OLS may actually be underestimating the variability for these parameters.

Both MEduc and FEduc change pretty significantly too, MEduc specifically. Mother's education standard error is much smaller than the OLS's, which means that there's more variation in the estimated Mother's education under the OLS. FEduc and Sibs increased a relative amount, which means that the OLS model fell short of accounting for the actual variabiulity that these covariates bring to model.

To add onto MEduc underestimate, variables such as Hours (lower by 7000) and TVHours (lower by 5000) were also lower in the Jackknife method than the OLS estimates. This means that there might be influential points that had inflated the OLS error estimates for these variables. This overall could have affected the outcome that the model had while testing the significance of the covariates.

Nonparametric Bootstrapping Method V.S. OLS Standard Error estimates

The results gotten from the nonparametric boostrapping method are about equal to the results that we got from the Jackknife method. All of the results from the nonparametric boostrapping method are within 1000 dollars of each other (excluding the intercept which is about 1500 away), which means they are so closely related that the results from the Jackknife VS OLS will apply to the nonparametric test results.

The fact that the Jackknife and Nonparametric Bootstrap got almost identical standard errors is actually quite a good sign for the validity of the project. The resampling results were good and stable provided evidence for each other by yielding the same results. Both methods showed that certain variables, like the intercept and MEduc, are extremely suspicious under the OLS, since they are very under valued in that model compared to the Bootstrapping and Jackknife methods.

iii) Use the bootstrap estimates to construct different types of 95 percent CIs for the regression coefficients.

```
r par(mfrow = c(1, 2)) hist(betaboot[, 1], "FD", freq = FALSE, xlab =
colnames(X)[1], main = NA) abline(v = beta[1], col = 2, lty = 2, lwd = 4)
hist(betaboot[, 2], "FD", freq = FALSE, xlab = colnames(X)[2], main = NA)
abline(v = beta[2], col = 2, lty = 2, lwd = 4)
```

Normal Confidence Interval

```
r alpha = 0.05 confint(full_model)
##                    2.5 %      97.5 % ## (Intercept) -1139787.136 1563672.818
## Age            -9282.145   15180.975 ## Educ          -49482.462   90379.388
## Hours         -20949.732    2302.771 ## TVHours      -148773.710   18047.486
## MEduc         -67900.262   60469.842 ## FEduc         -46177.946   66588.907
## Sibs           -7087.921  124385.860
r CInormal = cbind(betabar - qnorm(1 - alpha/2) * SE, betabar + qnorm(1 -
alpha/2) * SE) print(CInormal)
##             [,1]         [,2] ## [1,] -1364033.45 1798371.7354 ## [2,]
-10816.24   16551.2970 ## [3,]   -46362.84   87170.4796 ## [4,]   -19406.59
734.9154 ## [5,]  -135998.61    4129.9256 ## [6,]   -35352.96   27717.6132 ##
[7,]   -56649.55   76952.5088 ## [8,]   -14664.24  133740.8946
```

Percentile Confidence Interval

```
r CIpercentile = cbind(apply(betaboot, 2, quantile, probs = alpha/2),
apply(betaboot, 2, quantile, probs = 1 - alpha/2)) print(CIpercentile)
##             [,1]         [,2] ## [1,] -1471828.75 1715204.6428 ## [2,]
-10940.02   16948.9285 ## [3,]   -40216.35   92529.3238 ## [4,]   -20480.62
-416.9613 ## [5,]  -143808.24   -4352.3522 ## [6,]   -36998.65   26156.2166 ##
[7,]   -49608.70   84743.9113 ## [8,]    -6587.59  141444.0849
```

Pivotal Confidence Interval

```
r CIpivotal = cbind(2 * beta - apply(betaboot, 2, quantile, probs = 1 - alpha/2),
2 * beta - apply(betaboot, 2, quantile, probs = alpha/2)) print(CIpivotal)
##                    [,1]        [,2] ## (Intercept) -1291318.96 1895714.431 ##
Age           -11050.10   16838.853 ## Educ          -51632.40   81113.273 ##
Hours         -18230.00    1833.655 ## TVHours      -126373.87   13082.017 ##
MEduc         -33586.64   29568.233 ## FEduc         -64332.95   70019.661 ##
Sibs          -24146.15  123885.529
```

iv) How do the results of these methods compare against each other?

Above there are 4 different types of 95% confidence intervals using the non-parametric bootstrapping method for each regression coefficients, as well as a histogram showing normality in the intercept and Income (y -variable), separately. The confidence intervals utilized are a regular T-Test, CInormal, CIpercentile, and CIpivitol, all targeting slightly different parts of the linear model to help construct the confidence intervals.

For the most part the CInormal intervals are wider than the OLS interval with MEduc and Hours as exceptions and TVHours as exceptions because, as said in part ii, the OLS likely overestimated those variables significantly, making their confidence intervals wide and not too precise. This is because the OLS model is less reliable because it assumed the errors were all normal and homoscedastic (assuming it achieved all assumptions of the Gauss-Markov assumptions) and may have under-estimated most of the variables (including the intercept). When in reality that is not true due to the outliers and leverage points throughout the dataset.
It is important to draw attention to the Percentile CI for the variable Hours and TVHours. These are the ONLY intervals that does NOT contain 0, Hours: [-20480.62, -416.9613], and TVHours: [-143808.24, -4352.3522]. Which means that the pivitol CI, which corrects for bias in the boostrap distribution, calculated that Hours and TVHours are significant covariates. It is the only CI that marked Hours and TVHours as significant, since every other CI had Hours interval contain zero, which may mean that it is incorrect and fell victim to 5% likelihood of a type 1 error. But it could also mean that when adjusting for the specific skew or bias that the data has using the Pivitol method, these variables might actually be statistically significant. Which means we will have to continue to evaluate and test Hours and TVhours.
Overall, OLS, Normal, and Percentile's intervals all contained 0, so there was nothing statistically significant between them. Even if some intervals were wider or narrower than each other, they ultimately lead to the same conclusion in the end: failing to reject the null hypothesis for all variables at the 5% confidence level. But percentile shows that there may actually be some significance within the Hours and TVHours variables, giving a glimmer of hope for rejecting the null hypothesis.

Step 6:

i) Perform a nonparametric test of overall statistical significance.

ii) Perform a non-parametric test of statistical significance for the effect of a single covariate of your choice on the response variable.

iii) How do the estimated p-values compare against the p-values obtained under the theoretical null distribution?

---

i) Perform a nonparametric test of overall statistical significance.
Permutation F-test: Getting Permutation F-Statistic and corresponding P-Value

```
r alpha = 0.05 f = summary(full_model)$fstatistic[1] print(f)
##    value ## 1.176606
r pvalue = pf(f, p, n - p - 1, lower.tail = FALSE) print(pvalue)
##     value ## 0.3102288
r nperm = 10000 fperm = numeric(nperm) for (k in 1:nperm) { Yperm = sample(Y)
perm = lm(Yperm ~ Age + Educ + Hours + TVHours + MEduc + FEduc + Sibs, data =
gss) fperm[k] = summary(perm)$fstatistic[1] } pvalue = (1 + sum(fperm > f))/(1 +
nperm) print(pvalue)
## [1] 0.2986701
```

The estimated p-value obtained through the permutation F test 0.2986701 is roughly equal to the one obtained under the theoretical null distribution 0.3102288.

Provides the distribution with theoretical, observed, and empirical information using the entire full model.

```
r hist(fperm, "FD", freq = FALSE, main = NA, xlab = "Permutation F Test
Statistics") curve(df(x, p, n - p - 1), add = TRUE, lty = 2, lwd = 4) abline(v =
qf(1 - alpha, p, n - p - 1), col = 2, lty = 2, lwd = 4) abline(v =
quantile(fperm, 1 - alpha), col = 4, lty = 2, lwd = 4) abline(v = f, col = 7, lty
= 2, lwd = 4) legend("topright", c("Theoretical Distribution", "Theoretical
Quantile", "Empirical Quantile", "Observed Statistic"), col = c(1, 2, 4, 7), lty
= rep(2, 4), lwd = rep(4, 4))
```

The distribution of the permutation F test statistics almost perfectly agrees with the theoretical null distribution in this case.

ii) Perform a non-parametric test of statistical significance for the effect of a single covariate of your choice on the response variable.

Performing a nonparametric permutation t-test for the effect of Hours on Income

```
r t = summary(full_model)$coefficients["Hours", 3] print(t)
## [1] -1.574122
r pvalue = 2 * pt(abs(t), n - p - 1, lower.tail = FALSE) print(pvalue)
## [1] 0.1158518
r aux = lm(Income ~ Age + Educ + TVHours + MEduc + FEduc + Sibs, data = gss)
fitted = aux$fitted.values residuals = aux$residuals tperm = numeric(nperm) for
(k in 1:nperm) { Yperm = fitted + sample(residuals) perm = lm(Yperm ~ Age + Educ
+ Hours + TVHours + MEduc + FEduc + Sibs, data = gss) tperm[k] =
summary(perm)$coefficients["Hours", 3] } pvalue = (1 + sum(abs(tperm) >
abs(t)))/(1 + nperm) print(pvalue)
## [1] 0.1187881
```

The estimated p-value obtained through the permutation t-test 0.1187881 after utilizing the Hours covariate is roughly equal to the one obtained under the theoretical null distribution 0.1158518.

Permutation T-Test statistic: Provides the distribution with theoretical, observed, and empirical information using the Hours covariate.

```
r hist(tperm, "FD", freq = FALSE, main = NA, xlab = "Permutation t Test
Statistics") curve(dt(x, n - p - 1), add = TRUE, lty = 2, lwd = 2) abline(v =
qt(c(alpha/2, 1 - alpha/2), n - p - 1), col = 2, lty = 2, lwd = 2) abline(v =
quantile(tperm, c(alpha/2, 1 - alpha/2)), col = 4, lty = 2, lwd = 2) abline(v =
t, col = 7, lty = 2, lwd = 2) legend("topright", c("Theoretical Distribution",
"Theoretical Quantiles", "Empirical Quantiles", "Observed Statistic"), col = c(1,
2, 4, 7), lty = rep(2, 4), lwd = rep(2, 4), cex = 0.5)
```

The distribution of the permutation t test statistics almost perfectly agrees with the theoretical null distribution in this case.

Performing a nonparametric permutation t-test for the effect of TVHours on Income

```
r t = summary(full_model)$coefficients["TVHours", 3] print(t)
## [1] -1.538195
r pvalue = 2 * pt(abs(t), n - p - 1, lower.tail = FALSE) print(pvalue)
## [1] 0.1243934
```

```
r aux = lm(Income ~ Age + Educ + Hours + MEduc + FEduc + Sibs, data = gss) fitted
= aux$fitted.values residuals = aux$residuals tperm = numeric(nperm) for (k in
1:nperm) { Yperm = fitted + sample(residuals) perm = lm(Yperm ~ Age + Educ +
Hours + TVHours + MEduc + FEduc + Sibs, data = gss) tperm[k] =
summary(perm)$coefficients["TVHours", 3] } pvalue = (1 + sum(abs(tperm) >
abs(t)))/(1 + nperm) print(pvalue)
## [1] 0.1022898
```

The estimated p-value obtained through the permutation permutation t-test after utilizing the TVHours covariate 0.1022898 is slightly smaller than the one to the one obtained under the theoretical null distribution 0.1243934.

Permutation T-Test statistic: Provides the distribution with theoretical, observed, and empirical information using the TVHours covariate.

```
r hist(tperm, "FD", freq = FALSE, main = NA, xlab = "Permutation t Test
Statistics") curve(dt(x, n - p - 1), add = TRUE, lty = 2, lwd = 2) abline(v =
qt(c(alpha/2, 1 - alpha/2), n - p - 1), col = 2, lty = 2, lwd = 2) abline(v =
quantile(tperm, c(alpha/2, 1 - alpha/2)), col = 4, lty = 2, lwd = 2) abline(v =
t, col = 7, lty = 2, lwd = 2) legend("topright", c("Theoretical Distribution",
"Theoretical Quantiles", "Empirical Quantiles", "Observed Statistic"), col = c(1,
2, 4, 7), lty = rep(2, 4), lwd = rep(2, 4), cex = 0.5)
```

The distribution of the permutation t test statistics is slightly more concentrated than the theoretical null distribution in this case.

iii) How do the estimated p-values compare against the p-values obtained under the theoretical null distribution?

Permutation T Test: Hours covariate

The Permutation T Test statistic almost perfectly agrees with the theoretical null distrubtion and that the p-value compares very similarly to how the theoretical null distribution expects it to act. The p-value is not significant since the permutation t test almost perfectly follows the theoretical null distribution. So the results from the parametric classic t test are consistent with the conclusions from the permutation t test, in both cases we fail to reject the null that the coefficient for Hours equals zero.

Permutation F Test

A very similar story is told with the permutation F test. The histogram and the theoretical distribution are essentially equal, with little variation difference between them, and the overall significances and p-values of the permutation F test and the theoretical null distribution are quite reasonable and similar, both indicating that the model is not statistically significant. This agreement tells us that the assumptions of the F-test do not heavily affect our inference.

Permutation T Test: TVHours covariate

Interestingly enough, when testing the p-value of TVHours on Income using the permutation t test, we get to see a slightly different output than the other two. The histogram does not perfectly follow the theoretical distribution, being much more concentrated around zero than the t-distribution assumed, indicating some deviation from the assumptions of the parametric test and telling us that the OLS model assumption of normally distributed errors may not be best for the TVHours covariate. However, even though this test is showing a little evidence of significance, it is still not enough to actually be considered statistically significant at the 5% level, since the p-value is 0.1 So, the pivotal CI was not entirely accurate when determining these two covariates as statistically signficant, and we do not have sufficient evidence to conclude that TVhours has a statistically significant effect on income.

Step 7:

i) Are there any covariates which should be transformed?

ii) Should a transformation be applied to the response variable?

iii) Refit your linear model to account for all your transformed variables.

iv) Are the standardized residuals from your linear model normally distributed?

v) Construct 95 percent simultaneous confidence bands to test the global null hypothesis of normality for the standardized residuals.

---

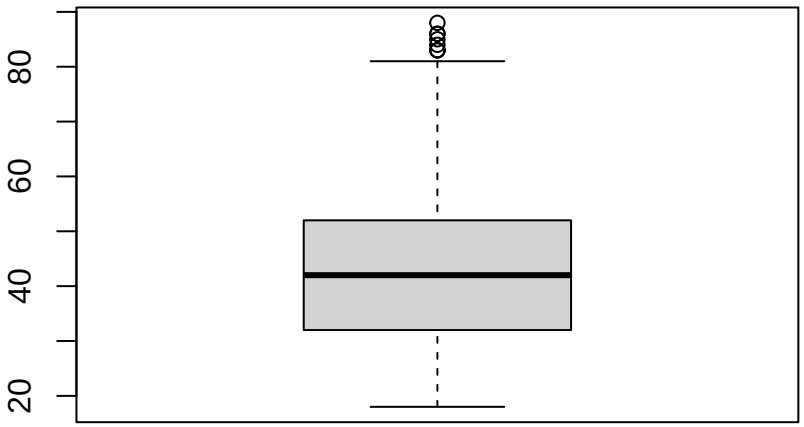
i) Are there any covariates which should be transformed?

We will look at each coefficient individually using a histogram and boxplot to look for skew.

Age

```
hist(gss$Age)
```

```r
boxplot(gss$Age)
```

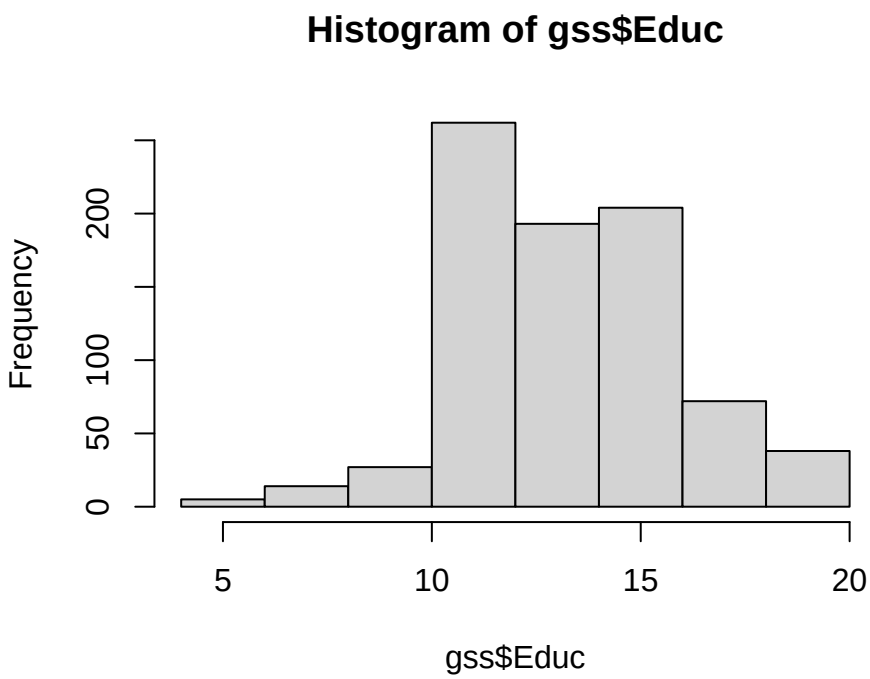


We can see that the histogram is roughly bell shaped with mild right skew and likewise the boxplot

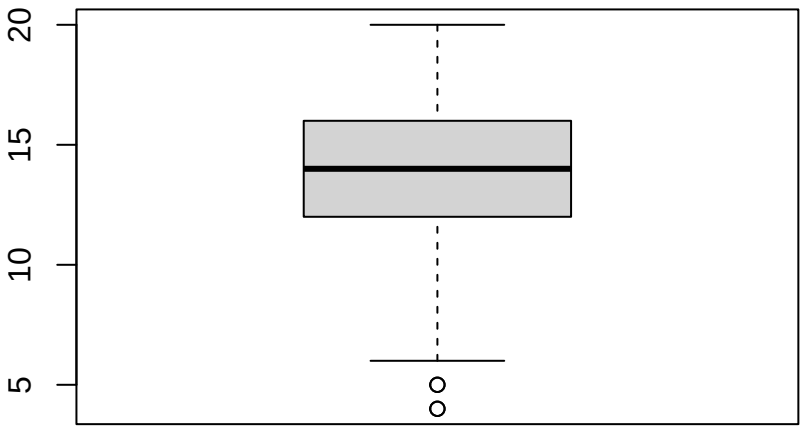
has a reasonable spread with some large outliers but not extreme enough to require a transformation.

Education

```
hist(gss$Educ)
```



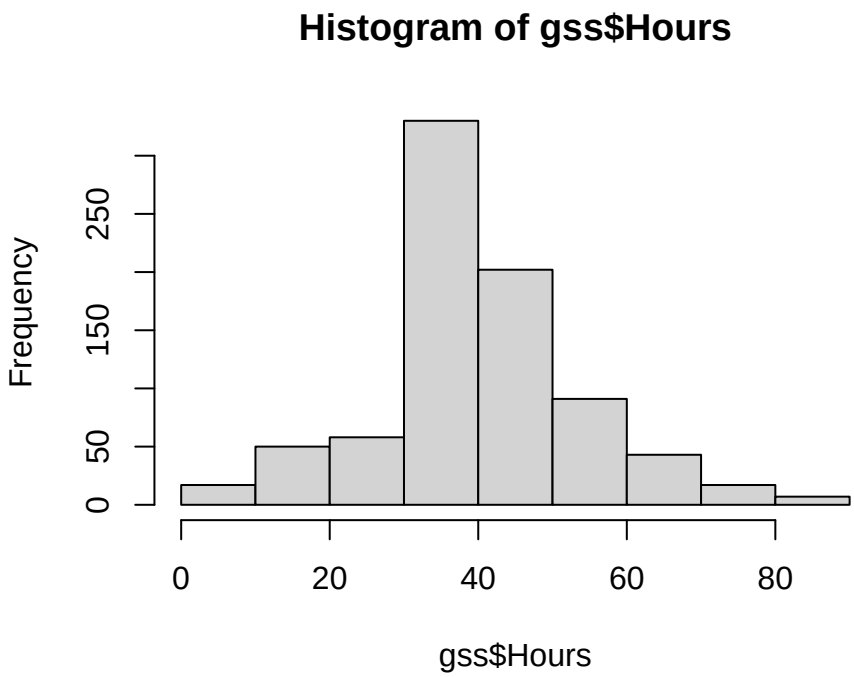

```
boxplot(gss$Educ)
```



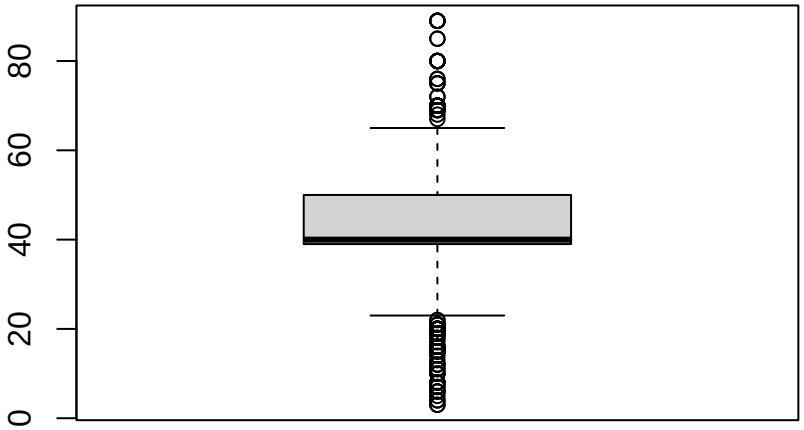
The histogram for education is roughly normal with a slight left tail and the boxplot also has a reasonable spread with some low outliers, but again not extreme enough to suggest a transformation.

Hours

```
hist(gss$Hours)
```




```r
boxplot(gss$Hours)
```

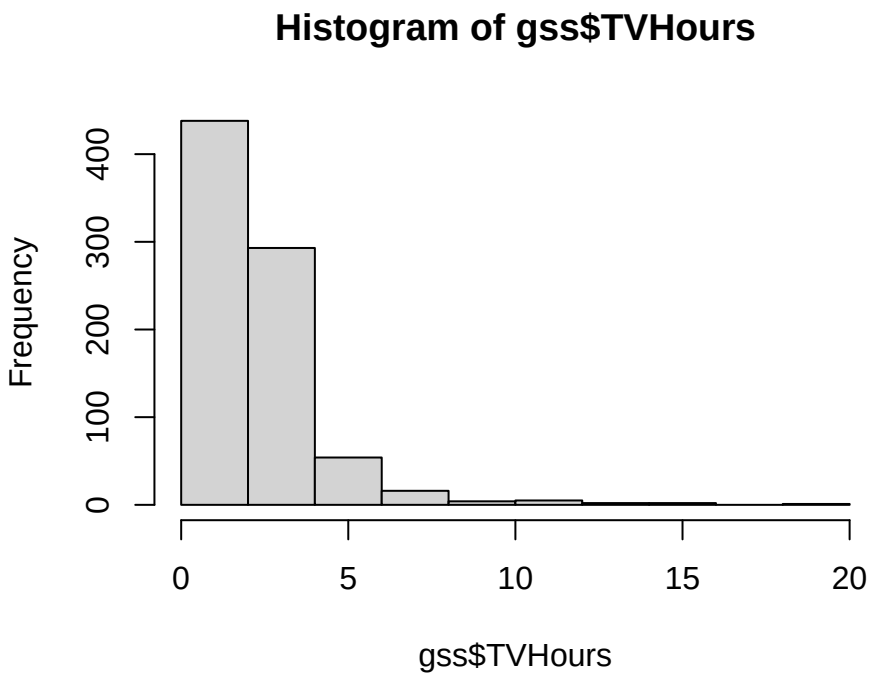


The histogram for hours is roughly symmetrical with some right skew. The boxplot for hours

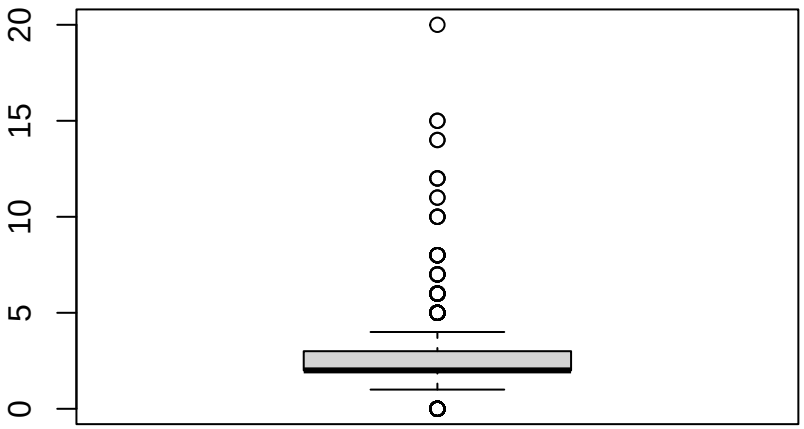
contains many outliers, but the outliers are on both the lower and upper ends so there is not evidence that a transformation would not improve anything.

TVHours

```
hist(gss$TVHours)
```



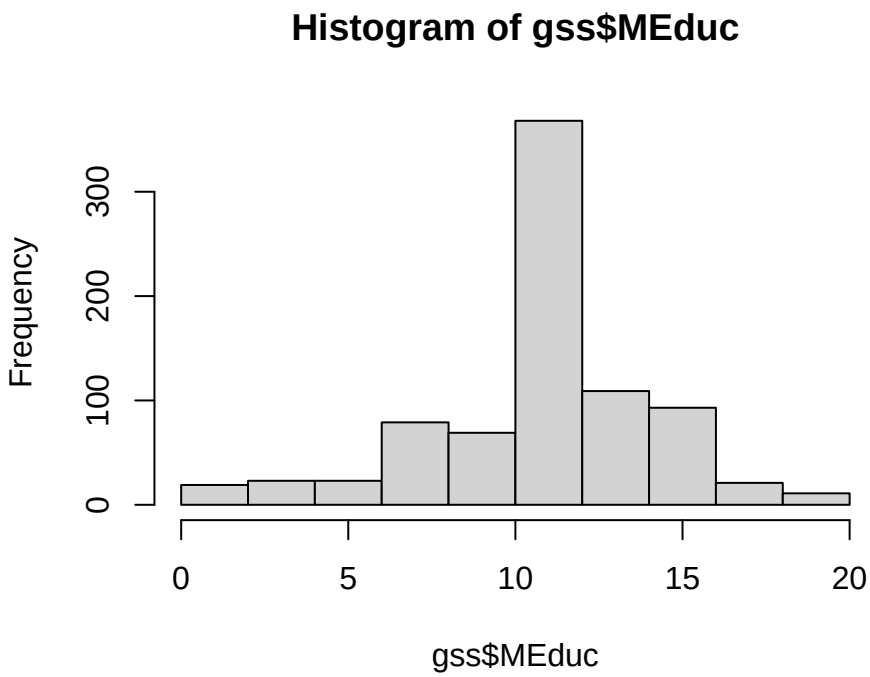

```
boxplot(gss$TVHours)
```



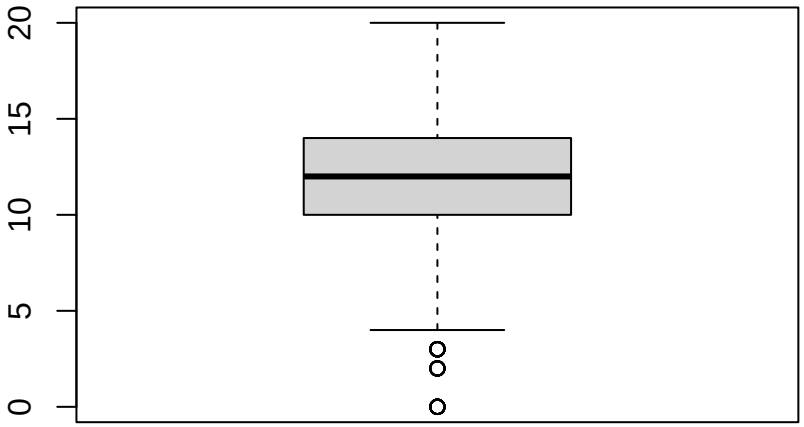
The histogram for TVHours is extremely right skewed with most values below 5 hours and a few very large values, and likewise the boxplot for TVHours also has severe skew and many high outliers, so a transformation is necessary. But since it includes 0 values, specifically a log transformation like log(TVHours + 1) could help reduce skew and improve variance for the TVHours covariate.

MEduc

```
hist(gss$MEduc)
```




```r
boxplot(gss$MEduc)
```

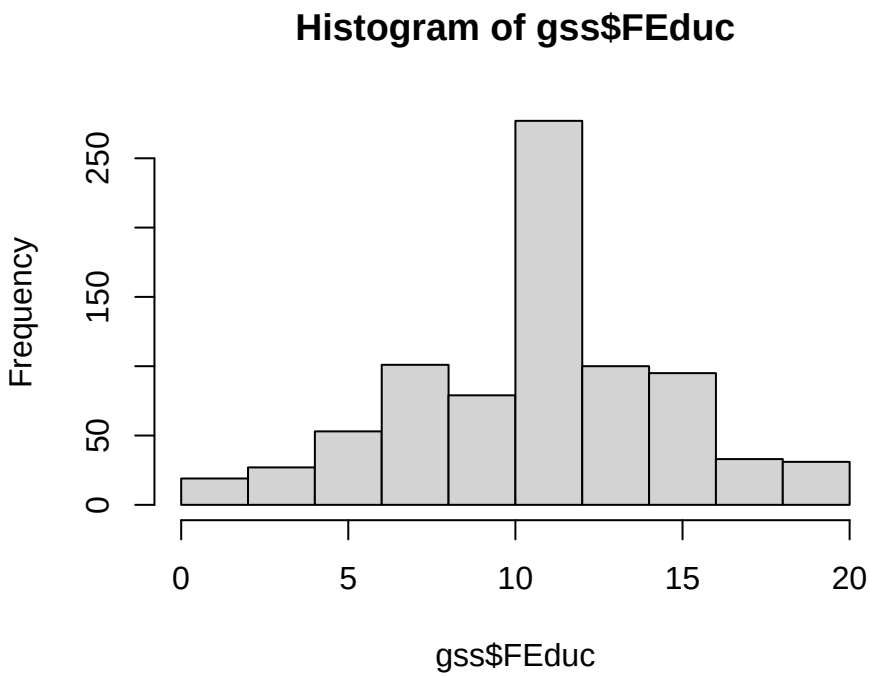


The histogram MEduc for mother's education is roughly symmetric with a very slight left tail. The

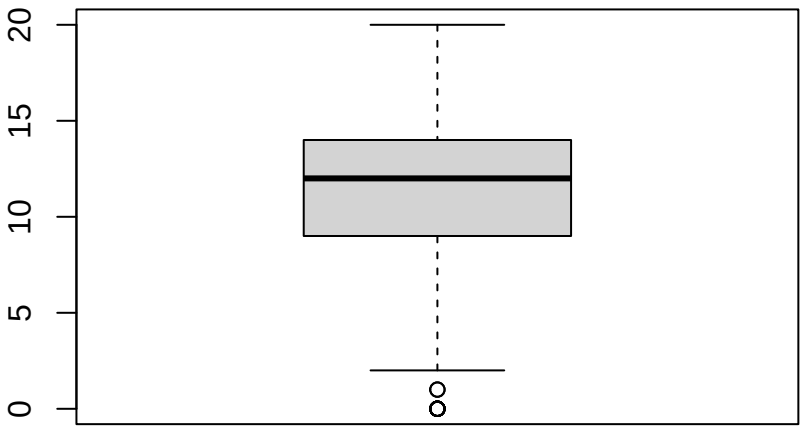
boxplot for MEduc is also reasonably spread with a few low outliers, so a transformation is not necessary.

FEduc

```
hist(gss$FEduc)
```




```
boxplot(gss$FEduc)
```

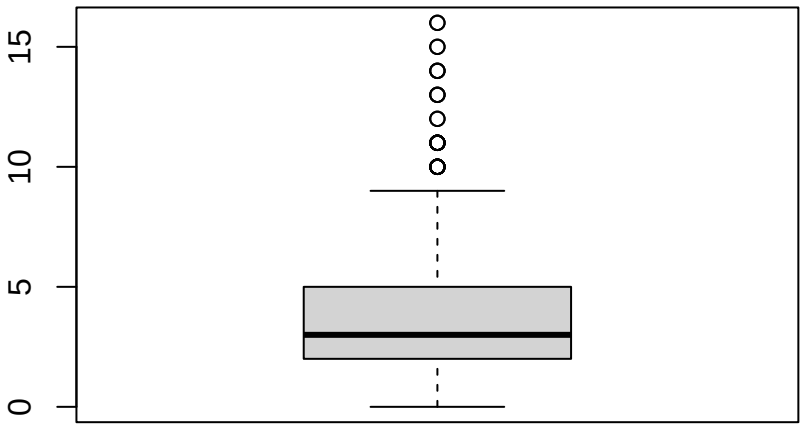


The histogram FEduc for father's education is also roughly symmetric with a very slight left tail, and the boxplot is reasonable with a couple low outliers. So a transformation is not needed for the FEduc covariate.

Sibs

```
hist(gss$Sibs)
```

```r
boxplot(gss$Sibs)
```



The histogram for number of siblings is strongly right skewed, and the box plot is also right skewed

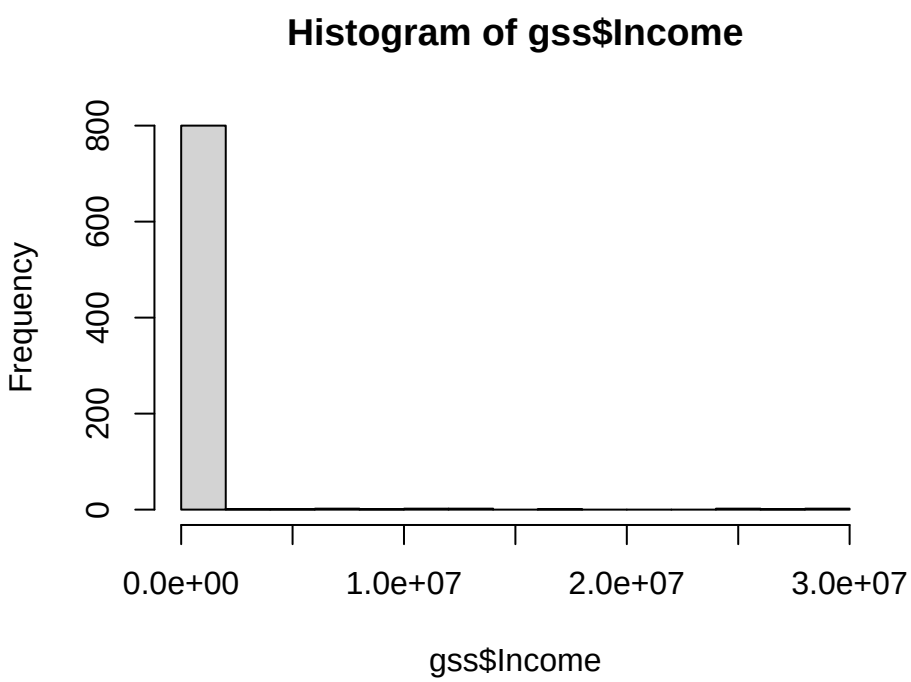
with several high outliers. Like the TVHours, though slightly less skeweed, the Sibs covariate includes 0 so it could use a transformation like log(Sibs + 1) to reduce variance and skewness.

Out of the 7 numerical covariates - Age (Responder's Age), Educ (Total years of education), Hours (Weekly work hours), TVHours (Hours spent watching TV daily), MEduc (Mother's years of education), FEduc (Father's years of education), and Sibs (Number of siblings)- TVHours and Sibs- are right skewed and to address their high values a log transformation is appropriate in order to improve normality and stabilize variance. While square root transformations are these covariates are substantially skewed and should be transformed with a log transformation to better satisfy model assumptions of normality and non constant variance.

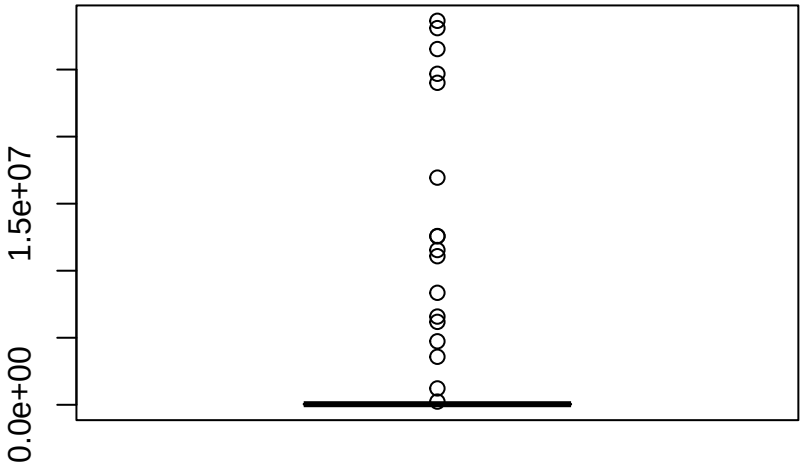
ii) Should a transformation be applied to the response variable?

Income

```
hist(gss$Income)
```



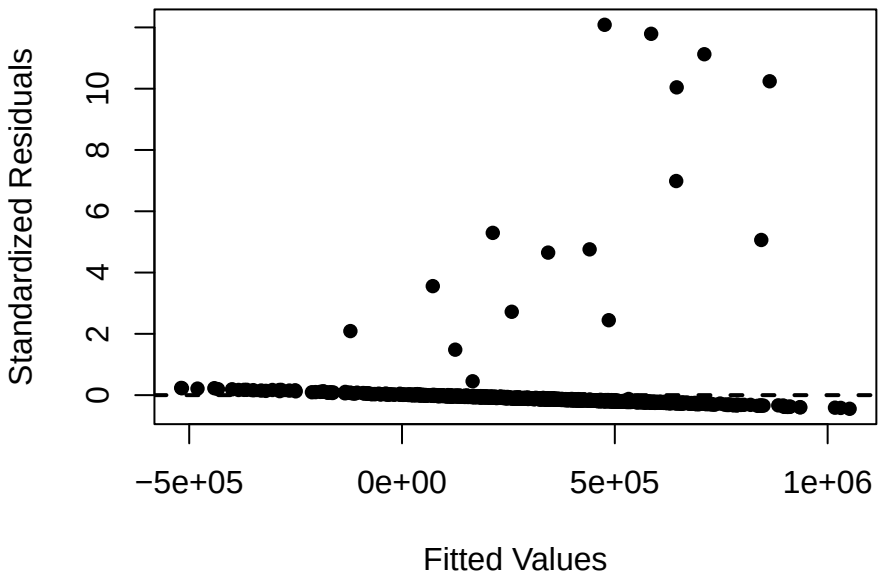

```
boxplot(gss$Income)
```



The histogram for Income as expected is extremely right skewed and the boxplot for income is nowhere near normal as the median is all the way at the bottom and there are many high outliers. Based on this evidence it is plausible that a transformation of the response is appropriate. We should refit the model using transformed covariates for TVHours and Sibs then re-evaluate residual diagnostics and if violations still persist we will consider transforming the response variable Income using a Box-Cox transformation to diagnose which transformation is appropriate.

Now we will plot the fitted values and standardized residuals to detect violations of the assumptions of linear regression.

```
fit = lm(Income ~ Age + Educ + Hours + log(TVHours + 1) + log(Sibs +
    1) + MEduc + FEduc, data = gss)
plot(fit$fitted.values, rstandard(fit), xlab = "Fitted Values", ylab = "Standardized Residuals"
    pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



After fitting the model with the transformed covariates, the residual plot still shows evidence for heteroscedasticity as the residuals are more concentrated around zero for smaller fitted values then we see large positive residuals at higher fitted values so residuals become more spread out as the fitted values increase, serving as an indicator for the violation of the constant variance assumption.

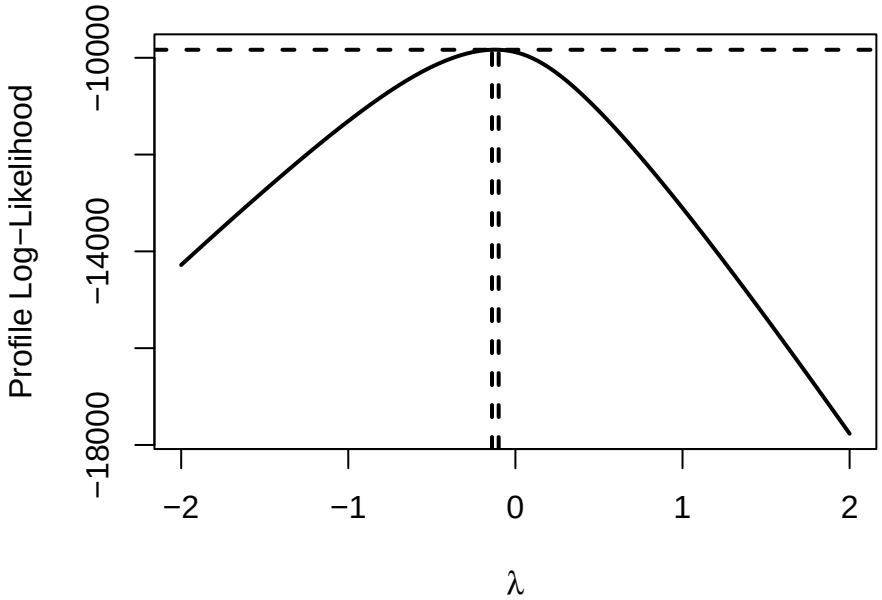
```
Y = gss$Income
alpha = 0.05
lambda = seq(-2, 2, 0.01)
loglik = numeric(length(lambda))
for (i in 1:length(lambda)) {
    if (lambda[i] == 0) {
        Ypower = log(Y)
    } else {
        Ypower = (Y^lambda[i] - 1)/lambda[i]
    }
    power = lm(Ypower ~ Age + Educ + Hours + log(TVHours + 1) + MEduc +
        FEduc + log(Sibs + 1), data = gss)

    loglik[i] = logLik(power)[1] + (lambda[i] - 1) * sum(log(Y))
}
```

```
CI = range(lambda[loglik > max(loglik) - qchisq(alpha, 1, lower.tail = FALSE)/2])
print(CI)
```

```
## [1] -0.14 -0.10
```

```r
plot(lambda, loglik, "l", xlab = expression(lambda), ylab = "Profile Log-Likelihood",
    lwd = 2)
abline(h = max(loglik) - qchisq(alpha, 1, lower.tail = FALSE)/2, lty = 2,
    lwd = 2)
abline(v = CI, lty = 2, lwd = 2)
```

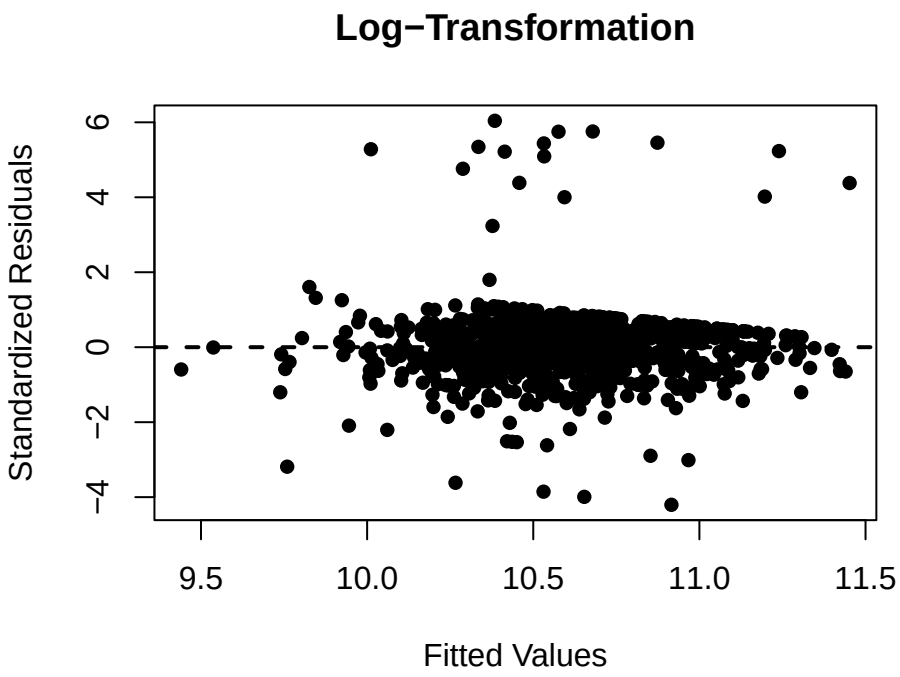


If 0 is inside the CI then a log transformation is used, but the CI is [-0.14 -0.10] which does not include 0 but is very close. Since our response variable Income was was very skewed and needs a transformation and the optimal lambda is close to 0, a log transformation is reasonable. We confirm that a transformation be applied to the response variable.

iii) Refit your linear model to account for all your transformed variables.

```r
fit_final = lm(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) +
    MEduc + FEduc + log(Sibs + 1), data = gss)
```

```r
plot(fit_final$fitted.values, rstandard(fit_final), main = "Log-Transformation",
    xlab = "Fitted Values", ylab = "Standardized Residuals", pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```

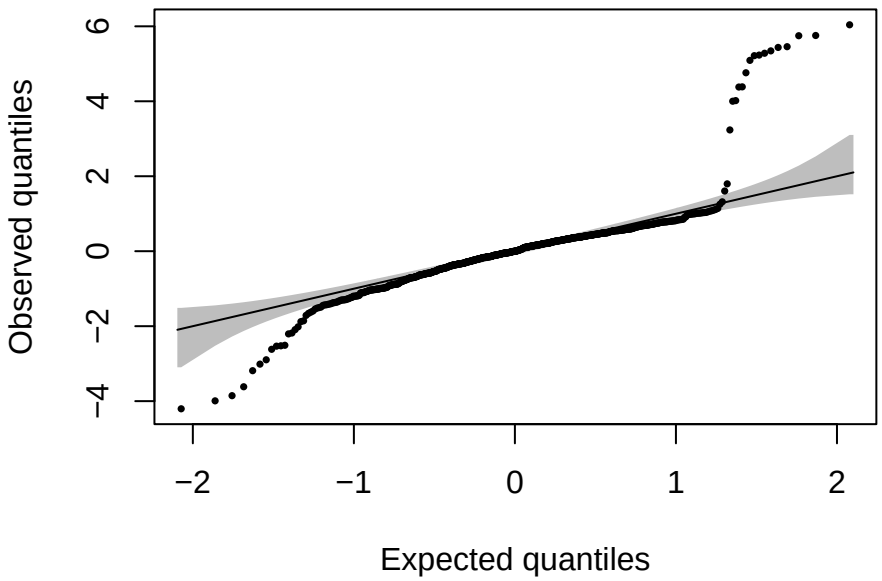


This plot looks much better as the residuals are centered around 0 and the spread is much more constant across the fitted values. We no longer see dramatic patterns and this transformation significantly improved the model assumptions. We do not have evidence to suggest a violation of the linearity of homoscedasticity assumptions.

iv) Are the standardized residuals from your linear model normally distributed?

```
library(qqconf)
qq_conf_plot(rstandard(fit_final), points_params = list(pch = 16, cex = 0.5))
```

```
## no dparams supplied. Estimating parameters from the data...
```



We use a Q-Q plot to display the observed quantiles of the sample against the expected quantiles of the null distribution. If the residuals were normally distributed we would expect the sample and theoretical quantiles to be in agreement and have the points form a straight line. We can see the points to not form a straight line at all as there is much deviation from the line at the lower and higher end of the quantiles. Thus we have evidence for a deviation from linearity, especially at the upper tail there is extreme upward curvature. This is consistent with our dataset as we found heavy right skeweness and extreme outliers. So no, the standardized residuals from the linear model are not normally distributed and there is evidence for the violation of the normality assumption.

v) Construct 95 percent simultaneous confidence bands to test the global null hypothesis of normality for the standardized residuals.

The plot above draws 95% simultaneous confidence bands around the Q-Q plot line. As all points on the Q-Q plot line do not lie within the simultaneous confidence bands, particularly the lower and upper tail points are far outside of the bands, so we reject the null hypothesis of normality at the 5% level.

Step 8:

Use your transformed variables from now on.

i) Investigate the interaction effects between the qualitative covariate which has been specified for your team and all available quantitative covariates on the response variable.

- What has been specified to us: Investigate the interaction effects between work status and all available quantitative covariates on the response variable.

ii) Formulate a suitable final model to draw inferences on the response variable and provide the interpretation of each type of coefficient in your final model.

---

i) Investigate the interaction effects between the 2 qualitative covariate which has been specified for your team and all available quantitative covariates on the response variable.

```r
full_int <- lm(log(Income) ~ Work * (Age + Educ + Hours + log(TVHours +
    1) + MEduc + FEduc + log(Sibs + 1)), data = gss)
summary(full_int)
```

```
##
## Call:
## lm(formula = log(Income) ~ Work * (Age + Educ + Hours + log(TVHours +
##     1) + MEduc + FEduc + log(Sibs + 1)), data = gss)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.7336 -0.4665  0.0148  0.4100  6.8045
##
## Coefficients: (2 not defined because of singularities)
##                                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)                       8.7815922  0.4684805  18.745  < 2e-16 ***
## WorkHousekeeping                 -1.7311362  1.5072929  -1.149  0.25112
## WorkLaid Off                      3.4539344  8.5312680   0.405  0.68570
## WorkNot Working                  -3.3742289  5.3420864  -0.632  0.52782
## WorkOther                        -5.6959180 11.6695883  -0.488  0.62562
## WorkParttime                      2.6917511  1.1564454   2.328  0.02020 *
## WorkRetired                       0.5030868  2.1874593   0.230  0.81816
## WorkSchool                        7.7889612  6.4313348   1.211  0.22624
## Age                               0.0135097  0.0044665   3.025  0.00257 **
## Educ                              0.0825263  0.0206934   3.988 7.31e-05 ***
## Hours                             0.0010146  0.0043255   0.235  0.81461
## log(TVHours + 1)                 -0.1947840  0.1094176  -1.780  0.07545 .
## MEduc                            -0.0082997  0.0191054  -0.434  0.66411
## FEduc                             0.0281617  0.0165624   1.700  0.08948 .
## log(Sibs + 1)                     0.1216572  0.0900660   1.351  0.17718
## WorkHousekeeping:Age              0.0007984  0.0126450   0.063  0.94967
## WorkLaid Off:Age                  0.0370831  0.0910906   0.407  0.68405
## WorkNot Working:Age              -0.1128065  0.1167466  -0.966  0.33423
```

```
## WorkOther:Age                      0.0135811  0.0766288   0.177  0.85937
## WorkParttime:Age                   0.0102467  0.0101200   1.013  0.31162
## WorkRetired:Age                   -0.0254000  0.0192396  -1.320  0.18717
## WorkSchool:Age                     0.0025342  0.0392249   0.065  0.94850
## WorkHousekeeping:Educ             -0.0427066  0.0686667  -0.622  0.53417
## WorkLaid Off:Educ                 -0.0993491  0.6328440  -0.157  0.87530
## WorkNot Working:Educ               0.5730530  0.4482570   1.278  0.20150
## WorkOther:Educ                     0.3171202  0.5450908   0.582  0.56089
## WorkParttime:Educ                 -0.1338918  0.0556635  -2.405  0.01640 *
## WorkRetired:Educ                   0.0776308  0.0864866   0.898  0.36968
## WorkSchool:Educ                   -0.2931145  0.1824101  -1.607  0.10850
## WorkHousekeeping:Hours             0.0119033  0.0110098   1.081  0.27997
## WorkLaid Off:Hours                -0.0279512  0.0752871  -0.371  0.71055
## WorkNot Working:Hours              0.0040528  0.0253795   0.160  0.87317
## WorkOther:Hours                    0.0144432  0.0995840   0.145  0.88472
## WorkParttime:Hours                -0.0064388  0.0105632  -0.610  0.54235
## WorkRetired:Hours                 -0.0008699  0.0120310  -0.072  0.94238
## WorkSchool:Hours                   0.0086353  0.0187365   0.461  0.64502
## WorkHousekeeping:log(TVHours + 1)  0.4225319  0.3085298   1.370  0.17125
## WorkLaid Off:log(TVHours + 1)      1.0348426  1.0411201   0.994  0.32056
## WorkNot Working:log(TVHours + 1)   1.6886198  1.5992931   1.056  0.29137
## WorkOther:log(TVHours + 1)         0.6526769  1.7689673   0.369  0.71226
## WorkParttime:log(TVHours + 1)     -0.7357416  0.2812336  -2.616  0.00907 **
## WorkRetired:log(TVHours + 1)       0.1809156  0.4012588   0.451  0.65221
## WorkSchool:log(TVHours + 1)       -0.9726081  0.6445197  -1.509  0.13171
## WorkHousekeeping:MEduc             0.0063575  0.0725264   0.088  0.93017
## WorkLaid Off:MEduc                -0.1799601  0.2625496  -0.685  0.49328
## WorkNot Working:MEduc             -0.1407431  0.2343068  -0.601  0.54823
## WorkOther:MEduc                   -0.0038677  0.3333820  -0.012  0.99075
## WorkParttime:MEduc                 0.0048837  0.0493252   0.099  0.92116
## WorkRetired:MEduc                 -0.0013848  0.0593096  -0.023  0.98138
## WorkSchool:MEduc                  -0.0460319  0.3050920  -0.151  0.88011
## WorkHousekeeping:FEduc             0.0536626  0.0617010   0.870  0.38473
## WorkLaid Off:FEduc                -0.0486919  0.2600907  -0.187  0.85155
## WorkNot Working:FEduc             -0.1823208  0.2305689  -0.791  0.42934
## WorkOther:FEduc                           NA         NA      NA       NA
## WorkParttime:FEduc                -0.0232042  0.0416333  -0.557  0.57746
## WorkRetired:FEduc                 -0.0485290  0.0566812  -0.856  0.39217
## WorkSchool:FEduc                  -0.2347872  0.2282919  -1.028  0.30407
## WorkHousekeeping:log(Sibs + 1)     0.0083198  0.2918398   0.029  0.97726
## WorkLaid Off:log(Sibs + 1)        -0.6295618  1.4804409  -0.425  0.67077
## WorkNot Working:log(Sibs + 1)      0.6286333  0.7795345   0.806  0.42025
## WorkOther:log(Sibs + 1)                   NA         NA      NA       NA
## WorkParttime:log(Sibs + 1)        -0.0605811  0.2444690  -0.248  0.80435
## WorkRetired:log(Sibs + 1)         -0.2720402  0.2997300  -0.908  0.36437
## WorkSchool:log(Sibs + 1)           0.0332938  0.5440693   0.061  0.95122
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 1.119 on 753 degrees of freedom
## Multiple R-squared:  0.1437, Adjusted R-squared:  0.07435
## F-statistic: 2.072 on 61 and 753 DF,  p-value: 7.01e-06
```

Work has 8 levels, so we have 7 dummy variables times our have 7 quantitative variables, so 49 interaction terms. We want to reduce our model to avoid over fitting. We perform backward elimination since we already have our linear model with all possible covariates and interactions.

ii) Formulate a suitable final model to draw inferences on the response variable and provide the interpretation of each type of coefficient in your final model.

We are using a backward elimination technique where we remove the lowest p-value per the ANOVA test, since we have quantitative and qualitative covariates.

```r
# anova(full_int)

full_int = update(full_int, . ~ . - Work:MEduc)
# anova(full_int)
full_int = update(full_int, . ~ . - Work:log(Sibs + 1))
# anova(full_int)
full_int = update(full_int, . ~ . - MEduc)
# anova(full_int)
full_int = update(full_int, . ~ . - Work:Hours)
# anova(full_int)
full_int = update(full_int, . ~ . - Work:FEduc)
# anova(full_int)
full_int = update(full_int, . ~ . - Work:Age)
# anova(full_int)
full_int = update(full_int, . ~ . - Hours)
# anova(full_int)
full_int = update(full_int, . ~ . - FEduc)
# anova(full_int)
full_int = update(full_int, . ~ . - Work:Educ)
# anova(full_int)
full_int = update(full_int, . ~ . - Work:log(TVHours + 1))
anova(full_int)
```

```
## Analysis of Variance Table
##
## Response: log(Income)
##                   Df Sum Sq Mean Sq F value    Pr(>F)
## Work               7  42.27   6.039  4.8832 2.065e-05 ***
## Age                1  17.53  17.532 14.1771 0.0001785 ***
## Educ               1  39.25  39.250 31.7387 2.440e-08 ***
## log(TVHours + 1)   1   8.16   8.159  6.5973 0.0103932 *
```

```
## log(Sibs + 1)      1   1.30   1.296  1.0483 0.3062043
## Residuals        803 993.03   1.237
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(full_int)
```

```
##
## Call:
## lm(formula = log(Income) ~ Work + Age + Educ + log(TVHours +
##     1) + log(Sibs + 1), data = gss)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.8316 -0.5004  0.0504  0.4226  6.7003
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)       9.241059   0.313054  29.519  < 2e-16 ***
## WorkHousekeeping -0.530434   0.158620  -3.344 0.000864 ***
## WorkLaid Off      0.039932   0.373731   0.107 0.914937
## WorkNot Working  -0.541399   0.312714  -1.731 0.083784 .
## WorkOther         0.077172   0.459982   0.168 0.866805
## WorkParttime     -0.139585   0.121542  -1.148 0.251124
## WorkRetired      -0.892310   0.201476  -4.429 1.08e-05 ***
## WorkSchool       -0.439255   0.316057  -1.390 0.164975
## Age               0.011528   0.003294   3.499 0.000492 ***
## Educ              0.082639   0.015980   5.171 2.94e-07 ***
## log(TVHours + 1) -0.225616   0.085992  -2.624 0.008863 **
## log(Sibs + 1)     0.072470   0.070781   1.024 0.306204
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.112 on 803 degrees of freedom
## Multiple R-squared:  0.09851,    Adjusted R-squared:  0.08616
## F-statistic: 7.977 on 11 and 803 DF,  p-value: 2.785e-13
```

After applying a backwards elimination technique on the transformed model, we get 12 predictors (including the intercept) as the lowest and best options for future predictions.

Inferences on the response variable: After applying a backwards elimination technique on the transformed model, we got a p-value of 2.785e-13, which means our new model is statistically significant while testing against Yearly Income. This means that these covariates actually influence the Income variable, to a significant degree. Though we did recieve an R^2 adj value of 0.08 which means that our variables are not explaining the variability of the model well. But, a significant p-value is a very good sign for this transformed model.

Interpretations of each type of coefficient: The Intercept Estimate (9.241059) This is the predicted log(Income) when all quantitative variables are zero and all qualitative variables are at their reference levels. In real terms, $e^{9.241059} = \$10,312$.

Variable: Educ (Estimate: 0.082639): For every additional year of education, a person's income is estimated to change by a factor of about $e^{0.082639} = 1.09$, or approximately a 9 increase, holding all other variables constant.

Variable: Age (Estimate: 0.011528) For every one-year increase in age, income is estimated to change by a factor of about $e^{0.011528} = 1.01$, or approximately a 1% increase, holding all other variables constant.

Variable: log(TVHours + 1) (Estimate: -0.225616) A 1% increase in the number of hours spent watching TV is associated with an estimated 0.23% decrease in income.

Variable: log(Sibs + 1) (Estimate: -0.072470) A 1% increase in the number of siblings one has is associated with an estimated 0.07% decrease in income.

Variable: WorkRetired (Estimate: -0.892310) For individuals who are retired, income is estimated to change by a factor of $e^{-0.892310} = 0.41$ (a decrease by approximately $59\%(1 - 0.41)$) relative to the typical person who works a full-time job and holding other factors constant. This sets a "new baseline income" prior to considering other factors of $\$4,228(10,312 * 0.41)$.

Variable: WorkHousekeeping (Estimate: -0.530434) For individuals who are housekeepers, income is estimated to change by a factor of $e^{-0.530434} = 0.59$ (a decrease by approximately $41\%(1 - 0.59)$) relative to the typical person who works a full-time job and holding other factors constant. This sets a "new baseline income" prior to considering other factors of $\$6,084(10,312 * 0.59)$.

Variable: WorkLaid Off (Estimate: 0.039932) For individuals who are laid off, income is estimated to change by a factor of $e^0.039932 = 1.04$ (an increase by approximately 4%) relative to the typical person who works a full-time job and holding other factors constant. This sets a "new baseline income" prior to considering other factors of $\$19,724(10,312 * 1.04)$.

Variable: WorkNot Working (Estimate: -0.541399) For individuals who are not working, income is estimated to change by a factor of $e^{-0.541399} = 1.72$ (an increase by approximately 72%) relative to the typical person who works a full-time job and holding other factors constant. This sets a "new baseline income" prior to considering other factors of $\$17,737(10,312 * 1.72)$.

Variable: WorkOther (Estimate: 0.077172) For individuals whose work is classified as "other," income is estimated to change by $e^0.077172 = 1.08$ (an increase by approximately 8%) relative to the typical person who works a full-time job and holding other factors constant. This sets a "new baseline income" prior to considering other factors of $\$11,137(10,312 * 1.08)$.

Variable: WorkParttime (Estimate: -0.139585) For individuals who work part time, income is estimated to change by a factor of $e^{-0.139585} = 0.87$ (a decrease by approximately 13%) relative to the typical person who works a full-time job and holding other factors constant. This sets a "new baseline income" prior to considering other factors of $\$8,971(10,312 * 0.87)$.

Variable: WorkSchool (Estimate: -0.439255) For individuals who work at a school, income is estiamted to change by a factor of $e^{-0.439255} = 0.64$ (a decrease by approximately 36%) relative to the typical person who works a full-time job and holding other factors constant. This sets a "new baseline income" prior to considering other factors of $\$6,600(10,312 * 0.64)$.

Step 9:

Ignore all interaction effects from now on.

i) Fit a mixed-effects model with all available quantitative and qualitative covariates by appropriately treating some of your covariates as random effects. Determine a suitable transformation for your response variable.

ii) Test the statistical significance of all your fixed effects.

iii) Calculate the intraclass correlation coefficients corresponding to each of your random effects What percentage of the total variation in your response variable does each of your random effects explain?

iv) What's another interpretation for each intraclass correlation coefficient?

---

i) Ignore all interaction effects from now on. Fit a mixed-effects model with all available quantitative and qualitative covariates by appropriately treating some of your covariates as random effects. Determine a suitable transformation for your response variable.

```r
library(lme4)

mixed_model = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) +
    Happy + MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 |
    Marital) + (1 | Work), data = gss)

sigma = data.frame(summary(mixed_model)$varcor)$sdcor
names(sigma) = data.frame(summary(mixed_model)$varcor)$grp
ICC = sigma^2/sum(sigma^2)
print(ICC)
```

```
##      Zodiac        Work     Marital    Residual
## 0.003463625 0.053966054 0.055065667 0.887504654
```

```r
summary(mixed_model, correlation = FALSE)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
##     MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 |
##     Marital) + (1 | Work)
##    Data: gss
##
## REML criterion at convergence: 2489.2
```

```
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.2700 -0.4054 -0.0073  0.3117  5.9940
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  Zodiac   (Intercept) 0.004464 0.06681
##  Work     (Intercept) 0.069556 0.26373
##  Marital  (Intercept) 0.070973 0.26641
##  Residual             1.143890 1.06953
## Number of obs: 815, groups:  Zodiac, 12; Work, 8; Marital, 5
##
## Fixed effects:
##                      Estimate Std. Error t value
## (Intercept)         8.2261484  0.4133664  19.900
## Age                 0.0090210  0.0034370   2.625
## Educ                0.0774983  0.0166455   4.656
## Hours               0.0003294  0.0031349   0.105
## log(TVHours + 1)   -0.1774335  0.0839616  -2.113
## HappyPretty Happy   0.3758767  0.1381257   2.721
## HappyVery Happy     0.5157427  0.1477088   3.492
## MEduc              -0.0105824  0.0150634  -0.703
## FEduc               0.0231915  0.0131725   1.761
## log(Sibs + 1)       0.1351602  0.0706531   1.913
## SexMale             0.1229452  0.0791305   1.554
```

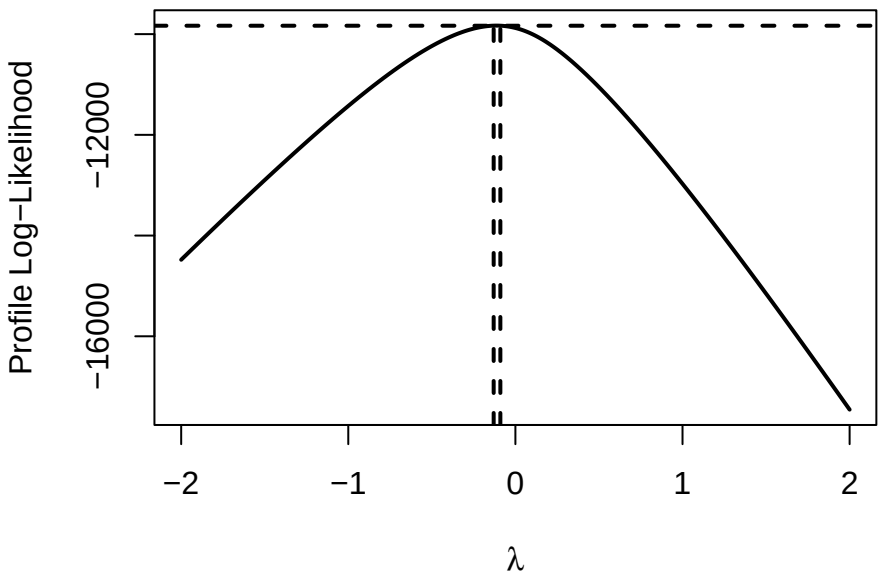
```r
Y = gss$Income
alpha = 0.05
lambda = seq(-2, 2, 0.01)
loglik = numeric(length(lambda))
for (i in 1:length(lambda)) {
    if (lambda[i] == 0) {
        Ypower = log(Y)
    } else {
        Ypower = (Y^lambda[i] - 1)/lambda[i]
    }
    power = lmer(Ypower ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
        MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
        (1 | Work), data = gss)

    loglik[i] = logLik(power)[1] + (lambda[i] - 1) * sum(log(Y))
}
```

```r
CI = range(lambda[loglik > max(loglik) - qchisq(alpha, 1, lower.tail = FALSE)/2])
print(CI)
```

```
## [1] -0.13 -0.09
```

```r
plot(lambda, loglik, "l", xlab = expression(lambda), ylab = "Profile Log-Likelihood",
    lwd = 2)
abline(h = max(loglik) - qchisq(alpha, 1, lower.tail = FALSE)/2, lty = 2,
    lwd = 2)
abline(v = CI, lty = 2, lwd = 2)
```



If 0 is inside the CI then a log transformation is used, but the CI is [-0.13 -0.09] which does not include 0 but is very close. Which also backsup our previous transformation that we did prior, showing that a log transformation IS a suitable transformation even for this mixed effects model. The residuals show a range from -4.27 to 5.99; the median is very close to zero (-0.0073) wihch suggests that the log transformation has done a good job of stabilizing the variance. So, we conclude that log(Income) is suitable even for this mixed effects model.

Since our response variable Income was was very skewed and needs a transformation and the optimal lambda is close to 0, a log transformation is reasonable. We confirm that

ii) Test the statistical significance of all your fixed effects.

Statistical Significance Testing for Age:

```r
full = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
reduced = lmer(log(Income) ~ Educ + Hours + log(TVHours + 1) + Happy +
```

```r
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)

LR = -2 * (logLik(reduced)[1] - logLik(full)[1])
pvalue = pchisq(LR, 5, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.2997707
```

The p-value of the Likelihood Ratio test for the effect of Age on log(Income) is equal to 0.2997707.

Statistical Significance Testing for Educ:

```r
full = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
reduced = lmer(log(Income) ~ Age + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, : Model failed
##   See ?lme4::convergence and ?lme4::troubleshooting.
```

```r
LR = -2 * (logLik(reduced)[1] - logLik(full)[1])
pvalue = pchisq(LR, 5, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.000610675
```

The p-value of the Likelihood Ratio test for the effect of Educ on log(Income) is equal to 0.000610675.

Statistical Significance Testing for Hours:

```r
full = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
reduced = lmer(log(Income) ~ Age + Educ + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)

LR = -2 * (logLik(reduced)[1] - logLik(full)[1])
pvalue = pchisq(LR, 5, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.9999998
```

The p-value of the Likelihood Ratio test for the effect of Hours on log(Income) is equal to 0.9999998

Statistical Significance Testing for log(TVHours + 1):

```
full = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
reduced = lmer(log(Income) ~ Age + Educ + Hours + Happy + MEduc + FEduc +
    log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) + (1 | Work),
    data = gss, REML = FALSE)

LR = -2 * (logLik(reduced)[1] - logLik(full)[1])
pvalue = pchisq(LR, 5, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.4771
```

The p-value of the Likelihood Ratio test for the effect of log(TVHours + 1) on log(Income) is equal to 0.4771

Statistical Significance Testing for Happy:

```
full = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
reduced = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)

LR = -2 * (logLik(reduced)[1] - logLik(full)[1])
pvalue = pchisq(LR, 5, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.0296004
```

The p-value of the Likelihood Ratio test for the effect of Happy on log(Income) is equal to 0.0296004.

Statistical Significance Testing for MEduc:

```
full = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
reduced = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) +
    Happy + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)

LR = -2 * (logLik(reduced)[1] - logLik(full)[1])
pvalue = pchisq(LR, 5, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.9921011
```

The p-value of the Likelihood Ratio test for the effect of MEduc on log(Income) is equal to 0.9921011

Statistical Significance Testing for FEduc:

```
full = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
reduced = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) +
    Happy + MEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)

LR = -2 * (logLik(reduced)[1] - logLik(full)[1])
pvalue = pchisq(LR, 5, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.6757152
```

The p-value of the Likelihood Ratio test for the effect of FEduc on log(Income) is equal to 0.6757152

Statistical Significance Testing for log(Sibs + 1):

```
full = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
reduced = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) +
    Happy + MEduc + FEduc + Sex + (1 | Zodiac) + (1 | Marital) + (1 |
    Work), data = gss, REML = FALSE)

LR = -2 * (logLik(reduced)[1] - logLik(full)[1])
pvalue = pchisq(LR, 5, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.5954642
```

The p-value of the Likelihood Ratio test for the effect of log(Sibs + 1) on log(Income) is equal to 0.5954642

Statistical Significance Testing for Sex:

```
full = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) + Happy +
    MEduc + FEduc + log(Sibs + 1) + Sex + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
reduced = lmer(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) +
    Happy + MEduc + FEduc + log(Sibs + 1) + (1 | Zodiac) + (1 | Marital) +
    (1 | Work), data = gss, REML = FALSE)
```

```r
LR = -2 * (logLik(reduced)[1] - logLik(full)[1])
pvalue = pchisq(LR, 5, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.7789858
```

The p-value of the Likelihood Ratio test for the effect of Sex on log(Income) is equal to 0.7789858

Out of every fixed effect covariate, the only statistically significant ones were Educ (p-value of 0.000610675) and Happy (p-value of 0.0296004). Every other fixed effect covariate remained statistically insignificant as their p-values through the Likelihood Ratio test were above the alpha threshold of 0.05.

iii) Calculate the intraclass correlation coefficients corresponding to each of your random effects. What percentage of the total variation in your response variable does each of your random effects explain?

Zodiac Intraclass Correlation Coefficient:

```r
ICC_Zodiac = (0.004464/(0.004464 + 0.069556 + 0.070973 + 1.14389))
ICC_Zodiac
```

```
## [1] 0.003463464
```

Zodiac explains roughly 0.35% of the total variation in log(Income), which is extremely, extremely small. It has very low similarities between observations within the same group. TTwo people with the same Zodiac status, the predicted correlation between their log-incomes is 0.0034.

Work Intraclass Correlation Coefficient:

```r
ICC_Sibs = (0.069556/(0.004464 + 0.069556 + 0.070973 + 1.14389))
ICC_Sibs
```

```
## [1] 0.05396611
```

Work explains roughly 5.4% of the total variation in log(Income), which is pretty small. It has low similarities between observations within the same group. Two people with the same Work status, the predicted correlation between their log-incomes is 0.054.

Marital Intraclass Correlation Coefficient:

```r
ICC_Marital = (0.070973/(0.004464 + 0.069556 + 0.070973 + 1.14389))
ICC_Marital
```

```
## [1] 0.05506551
```

Marital explains roughly 5.5% of the total variation in log(Income), which is pretty small. It has low similarities between observations within the same group. Two people with the same marital status, the predicted correlation between their log-incomes is 0.055.

Residual Intraclass Correlation Coefficient:

```
ICC_Marital = (1.14389/(0.004464 + 0.069556 + 0.070973 + 1.14389))
ICC_Marital
```

```
## [1] 0.8875049
```

The residuals explain 88.75% of the total varition in log(Income), this means most of the total variation in log(Income) can not be explained by our random variables. Which is not optimal for our linear model, since we would rather have a higher degree of total variability in log(Income) be explained by our covaraites rather than just residuals.

iv) What's another interpretation for each intraclass correlation coefficient?

The ICC represents the expected correlation between log(Income) of two different individuals who belong to the same group, examples stated above in section iii just above. ICC is a measure of within group similarity. A higher ICC means that individuals within the same category are more alike regarding their income than individuals chosen from the population at large. And we do not particullary have any large ICC's for our variables, which indicates that there may not be a lot of simularity between our random groups.

Step 10:

Ignore all random effects from now on.

i) Apply a suitable model selection method and draw valid inferences on the response variable.

i) Apply a suitable model selection method and draw valid inferences on the response variable.

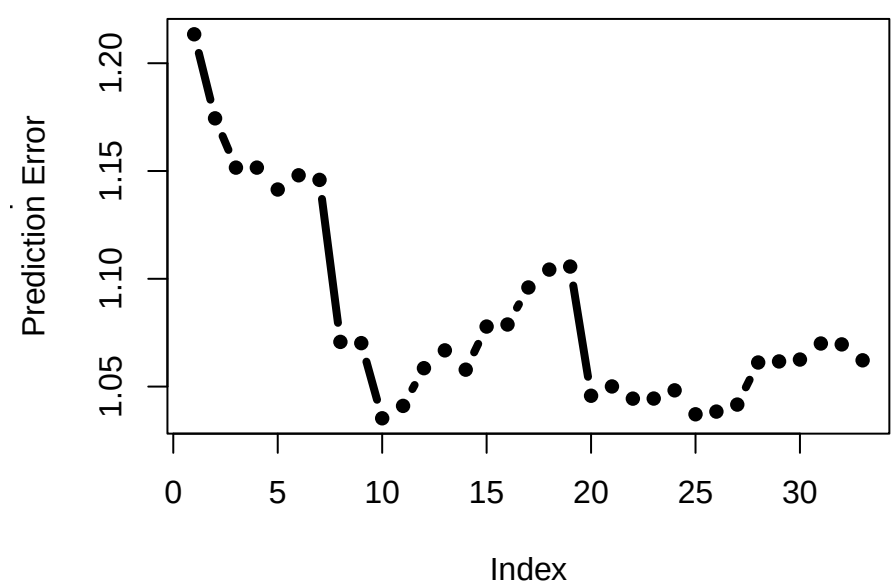
Using a Backward Elimination with Valid Post-Selection Inference to formulate a good final model to draw inferences on Income. A 60 | 20 | 20 stadnard validation split (60 for training 20 for validation 20 for testing).

```
set.seed(0)
Y = log(gss$Income)
gss_no_income <- gss[, names(gss) != "Income"]
X = as.data.frame(model.matrix(~. - 1, data = gss_no_income))
n = dim(X)[1]
```

```r
P = dim(X)[2]

train = sample(n, floor(0.6 * n))
valid = sample(setdiff(1:n, train), floor(0.2 * n))
test = setdiff(1:n, c(train, valid))
MSPE = numeric(P)

for (p in P:1) {
    fit = lm(Y ~ ., X[, 1:p, drop = FALSE], train)
    predictions = predict(fit, X[valid, 1:p, drop = FALSE])
    MSPE[p] = mean((Y[valid] - predictions)^2)
    ind = which.max(summary(fit)$coefficients[-1, 4])
    if (ind != p) {
        X[, c(ind, p)] = X[, c(p, ind)]
        names(X)[c(ind, p)] = names(X)[c(p, ind)]
    }
}
plot(MSPE, type = "b", pch = 16, lwd = 4, ylab = "Mean Squared
Prediction Error")
```




```r
p = which.min(MSPE)
fit = lm(Y ~ ., X[, 1:p, drop = FALSE], test)
summary(fit)
```

```
##
```

```
## Call:
## lm(formula = Y ~ ., data = X[, 1:p, drop = FALSE], subset = test)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.0268 -0.4040 -0.0029  0.3614  5.1584
##
## Coefficients: (1 not defined because of singularities)
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)              9.052408   0.586905  15.424  < 2e-16 ***
## Educ                     0.111808   0.034833   3.210  0.00162 **
## TVHours                 -0.088305   0.041668  -2.119  0.03568 *
## SexFemale                0.051140   0.162089   0.316  0.75281
## SexMale                        NA         NA      NA       NA
## Hours                   -0.009908   0.005543  -1.788  0.07581 .
## FEduc                    0.036970   0.023210   1.593  0.11325
## `WorkNot Working`       -0.772929   0.591049  -1.308  0.19293
## `MaritalNever Married` -0.007080   0.269710  -0.026  0.97909
## `HappyPretty Happy`      0.020939   0.161821   0.129  0.89721
## MaritalMarried           0.431851   0.229985   1.878  0.06232 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9873 on 153 degrees of freedom
## Multiple R-squared:  0.2122, Adjusted R-squared:  0.1659
## F-statistic:  4.58 on 9 and 153 DF,  p-value: 2.318e-05
```

Analysis: After applying the Backward elimination with valid post selection inference, we were able to get the maximum amount of covariates that our model should consist on in order to have the lowest predicition error: 10. This is clear from the graph that shows 10 covariates are at the lowest prediction error of around 1. After applying the backward elimination method, we were able to get two signficant p-values in TvHours and Educ. And to further, the overall p-value of the fitted model came out to 2.2e-16 which is approximately zero, showing that the model is statistically significant and that we can reject the null hypothesis. A nice change of pace from what our previous findings have been (which is having no statistically significant values). Also, our R^2 value was 0.166 which is a significant improvement from any model before, showing that our model actually explains 16.6% of the variation in the dataset. This model is by the best model that we have had up until this point, demonstrating that our corrections to certain problems with our data have been going well and proper.

Step 11:

i) Investigate the collinearity among all quantitative covariates.

ii) Fit different types of regularized regression models with all available quantitative covariates. What can you observe about the estimated regularized regression coefficients?

iii)

Apply the bootstrap method to obtain estimates for the standard errors of the estimated regularized regression coefficients. How do they compare against the estimates obtained through the ordinary least squares method?

i) Investigate the collinearity among all quantitative covariates.

In terms of correlation, there seems to be reasonable diversity within how correlated the quantiative covariates are with each other, nothing being above 0.75, so pairwise correlations are not too significant or impactful but are relatively moderate. The strongest pairwise correlation is between MEduc and FEduc with correlation coefficient 0.659. In terms of investigating collinearity of the quantiative covariates it is clear through the Variance Inflation Factor (VIF) values based on the final model that uses transformed TVHours and Sibs that there is no collinearity between the variables. All of the VIF values are between (1, 2), and a rule of thumb is that VIF values above 5 can indicate collinearity. The largest VIFs being MEduc and FEduc, 1.9 and 1.95 respecitevly, so there is no evidence of serious collinearity among the quantitative covariates in the model.

```r
gss_numeric <- gss[, c("Age", "Educ", "Hours", "TVHours", "MEduc", "FEduc",
    "Sibs")]
cor(gss_numeric)
```

```
##                 Age        Educ       Hours     TVHours       MEduc       FEduc
## Age      1.00000000 -0.04082701  0.00790323  0.10378955 -0.31815628 -0.35234245
## Educ    -0.04082701  1.00000000  0.12672660 -0.20078493  0.38880229  0.36922630
## Hours    0.00790323  0.12672660  1.00000000 -0.08477884  0.03961121  0.02398382
## TVHours  0.10378955 -0.20078493 -0.08477884  1.00000000 -0.15919591 -0.17010512
## MEduc   -0.31815628  0.38880229  0.03961121 -0.15919591  1.00000000  0.65907807
## FEduc   -0.35234245  0.36922630  0.02398382 -0.17010512  0.65907807  1.00000000
## Sibs     0.14790176 -0.27897107 -0.09786384  0.12701586 -0.30904693 -0.34790739
##                Sibs
## Age      0.14790176
## Educ    -0.27897107
## Hours   -0.09786384
## TVHours  0.12701586
## MEduc   -0.30904693
## FEduc   -0.34790739
## Sibs     1.00000000
```

```r
# Checking how correlated the quantitative covariates are to assist
# in finding the collinearity.

library(faraway)
vif(fit_final)
```

```
##              Age             Educ            Hours log(TVHours + 1)
##         1.190042         1.304091         1.025806         1.076900
##            MEduc            FEduc    log(Sibs + 1)
##         1.901266         1.953671         1.160950
```

ii) Fit different types of regularized regression models with all available quantitative covariates

Full OLS model coefficients

```r
coef(full_model)
```

```
## (Intercept)         Age        Educ       Hours     TVHours       MEduc
##  211942.841    2949.415   20448.463   -9323.480  -65363.112   -3715.210
##       FEduc        Sibs
##   10205.481   58648.969
```

```r
Y = gss$Income
X = gss_numeric
n = dim(X)[1]
P = dim(X)[2]
set.seed(4030)
train = sample(n, 0.6 * n)
valid = sample(setdiff(1:n, train), 0.2 * n)
test = setdiff(1:n, c(train, valid))
```

Ridge Regression:

```r
library(glmnet)
train = c(train, valid)
X = as.matrix(X)
cv.ridge = cv.glmnet(X[train, ], Y[train], nfolds = 5, alpha = 0)
ridge = glmnet(X[test, ], Y[test], alpha = 0, lambda = cv.ridge$lambda.min)
print(ridge$beta)
```

```
## 7 x 1 sparse Matrix of class "dgCMatrix"
##                 s0
## Age      -1273.665
## Educ     20129.530
## Hours    -1030.022
## TVHours -10473.137
## MEduc     1980.115
## FEduc    -8107.498
## Sibs     24759.200
```

The estimates for the estimated ridge regression coefficients are all much smaller than the corresponding ones in the full model, owing to the imposed penalty on the 2-norm of the regression coefficient vector. All estimated ridge regression coefficients appear to be slightly shrunk towards zero as a result. The coefficient for Age decreases from 2949.415 to -1273.665 with a sign change. The coefficient for Educ decreases slightly from 20448.463 to 20129.530. The coefficient for Hours moves closer to zero from -9323.480 to -1030.022. The coefficient for TVHours moves closer to zero from -65363.112 to -10473.137. The coefficient fror MEduc moves closer to zero from -3715.210 to 1980.115 with a sign change. The coefficient for FEduc moves closer to zero with a sign change from 10205.481 to -8107.498. Finally, the coefficient for Sibs decreases from 58648.969 to 24759.200.

Lasso Regression:

```r
cv.lasso = cv.glmnet(X[train, ], Y[train], nfolds = 5, alpha = 1)
lasso = glmnet(X[test, ], Y[test], alpha = 1, lambda = cv.lasso$lambda.min)
print(lasso$beta)
```

```
## 7 x 1 sparse Matrix of class "dgCMatrix"
##                 s0
## Age          .
## Educ     50283.021
## Hours        .
## TVHours      .
## MEduc        .
## FEduc    -5796.337
## Sibs     62643.151
```

We observe that the estimated lasso regression coefficients for Age, Hours, TVHours, and MEduc have been set to zero, so lasso removed these covariates from the model. The estimated lasso regression coefficient for FEduc shrunk closer to zero and changes sign, but the coefficients for Educ and Sibs increased.

iii) Apply the bootstrap method to obtain estimates for the standard errors of the estimated regularized regression coefficients.

```r
summary(full_model)$coefficients[-1, "Std. Error"]
```

```
##       Age      Educ     Hours   TVHours     MEduc     FEduc      Sibs
##  6231.347 35626.187  5922.973 42493.382 32698.962 28724.438 33489.543
```

Ridge Regression:

```r
nboot = 10000
beta_ridge = matrix(0, nboot, P)
colnames(beta_ridge) = colnames(X)
for (i in 1:nboot) {
    ind = sample(test, replace = TRUE)
    beta_ridge[i, ] = glmnet(X[ind, ], Y[ind], alpha = 0, lambda = cv.ridge$lambda.min)$beta[,
        ]
}
SE_ridge = apply(beta_ridge, 2, sd)
print(SE_ridge)
```

```
##       Age      Educ     Hours   TVHours     MEduc     FEduc      Sibs
##  1882.244 28629.840  1790.363 14139.763  3297.963 12401.428 35203.935
```

The bootstrap estimates for the standard errors of the estimated ridge regression coefficients generally are much smaller (besides Sibs) than the corresponding ones in the full model owing to the imposed penalty on the 2-norm of the regression coefficient vector. The standard error for Age decreases from 6231.347 to 1882.244. The standard error for Educ decreases from 35626.187 to 28629.840. The standard error for Hours decreases from 5922.973 to 1790.363. The standard error for TVHours decreases from 42493.382 to 14139.763. The standard error fror MEduc decreases from 32698.962 to 3297.963. The standard error for FEduc decreases from 28724.438 to 12401.428. Finally, the standard error for Sibs increases slightly from 33489.543 to 35203.935.

Lasso Regression:

```r
beta_lasso = matrix(0, nboot, P)
colnames(beta_lasso) = colnames(X)
for (i in 1:nboot) {
    ind = sample(test, replace = TRUE)
    beta_lasso[i, ] = glmnet(X[ind, ], Y[ind], alpha = 1, lambda = cv.lasso$lambda.min)$beta[,
        ]
}
SE_lasso = apply(beta_lasso, 2, sd)
print(SE_lasso)
```

```
##       Age      Educ     Hours   TVHours     MEduc     FEduc      Sibs
##  3898.264 86464.824  5104.711 14890.902 10259.079 33426.613 84808.710
```

The bootstrap estimates for the standard errors of the estimated lasso regression coefficients change in different ways to corresponding ones in the full model. The standard error for Age decreases from 6231.347 to 3898.264. The standard error for Educ increases from 35626.187 to 86464.824. The standard error for Hours decreases from 5922.973 to 5104.711. The standard error for TVHours decreases from 42493.382 to 14890.902. The standard error for MEduc decreases from 32698.962 to 10259.079. The standard error for FEduc increases from 28724.438 to 33426.613. Finally, the standard error for Sibs increases from 33489.543 to 84808.710.

Step 12:

i) Apply a suitable multiple comparisons method to compare the average response value across different levels defined by the qualitative covariate which has been specified for your team at 95 percent global confidence level.

ii) How do the results of this method compare against the results obtained without the application of the multiple comparisons method?

---

i) Apply a suitable multiple comparisons method to compare the average response value across different levels at a 95% gloval confidence level

We use Scheffe's and Tukey's method to construct confidence intervals at the 95 percent global confidence level. We do not want separate individual 95% CIs we want to invoke the family-wise error rate (FWER) to construct simultaneous CIs such that FWER is less than or equal to alpha (0.05).

Set up test for differences between average response values across all different work status pairs

```
fit = lm(Income ~ Work, gss)
alpha = 0.05
Y = gss$Income
Q = factor(gss$Work)
n = length(Q)
K = length(levels(Q))
```

Mean comparisons with no correction

```
mu = aggregate(Income ~ Work, gss, mean)[, 2]
S = sqrt(sum((Y - mu[Q])^2)/(n - K))
SE = S * sqrt(2 * K/n)
diff = outer(mu, mu, "-")
diff = diff[upper.tri(diff)]
lab = outer(levels(Q), levels(Q), function(x, y) {
    paste0(x, " - ", y)
})
names(diff) = lab[upper.tri(lab)]
CI = cbind(diff - qt(1 - alpha/2, n - K) * SE, diff + qt(1 - alpha/2,
    n - K) * SE)
colnames(CI) = c("Lower", "Upper")
print(CI)
```

```
##                                  Lower      Upper
## Fulltime - Housekeeping     -354140.92  935427.33
## Fulltime - Laid Off         -370191.30  919376.95
## Housekeeping - Laid Off     -660834.50  628733.75
## Fulltime - Not Working      -348862.24  940706.01
## Housekeeping - Not Working  -639505.44  650062.81
## Laid Off - Not Working      -623455.06  666113.18
## Fulltime - Other            -354913.52  934654.73
## Housekeeping - Other        -645556.72  644011.52
## Laid Off - Other            -629506.35  660061.90
## Not Working - Other         -650835.41  638732.84
## Fulltime - Parttime        -1033058.50  256509.75
## Housekeeping - Parttime    -1323701.70  -34133.46
## Laid Off - Parttime        -1307651.33  -18083.08
## Not Working - Parttime     -1328980.39  -39412.14
## Other - Parttime           -1322929.10  -33360.86
## Fulltime - Retired          -342812.07  946756.18
## Housekeeping - Retired      -633455.27  656112.97
## Laid Off - Retired          -617404.90  672163.35
## Not Working - Retired       -638733.96  650834.29
## Other - Retired             -632682.67  656885.57
## Parttime - Retired            45462.31 1335030.55
## Fulltime - School           -349227.62  940340.62
## Housekeeping - School       -639870.82  649697.42
## Laid Off - School           -623820.45  665747.80
## Not Working - School        -645149.51  644418.74
## Other - School              -639098.23  650470.02
## Parttime - School             39046.75 1328615.00
## Retired - School            -651199.68  638368.57
```

```r
library(emmeans)
confint(pairs(emmeans(fit, ~Work), adjust = "none"))
```

```
##  contrast                   estimate      SE  df lower.CL upper.CL
##  Fulltime - Housekeeping      290643  321000 807  -338711   919997
##  Fulltime - Laid Off          274593  788000 807 -1271422  1820608
##  Fulltime - Not Working       295922  658000 807  -994876  1586720
##  Fulltime - Other             289871  962000 807 -1598692  2178433
##  Fulltime - Parttime         -388274  255000 807  -889336   112787
##  Fulltime - Retired           301972  359000 807  -403382  1007326
##  Fulltime - School            295556  658000 807  -995242  1586355
##  Housekeeping - Laid Off      -16050  839000 807 -1662848  1630748
##  Housekeeping - Not Working     5279  718000 807 -1404665  1415222
##  Housekeeping - Other           -773 1000000 807 -1972688  1971142
##  Housekeeping - Parttime     -678918  386000 807 -1435782    77947
##  Housekeeping - Retired        11329  461000 807  -893826   916484
##  Housekeeping - School          4913  718000 807 -1405030  1414857
```

```
##  Laid Off - Not Working        21329 1020000 807 -1974170  2016828
##  Laid Off - Other              15278 1240000 807 -2410114  2440670
##  Laid Off - Parttime         -662867  816000 807 -2265028   939294
##  Laid Off - Retired            27379  855000 807 -1649935  1704693
##  Laid Off - School             20964 1020000 807 -1974536  2016463
##  Not Working - Other           -6051 1160000 807 -2277289  2265187
##  Not Working - Parttime      -684196  692000 807 -2041737   673344
##  Not Working - Retired          6050  736000 807 -1439418  1451518
##  Not Working - School           -365  920000 807 -1805362  1804632
##  Other - Parttime            -678145  986000 807 -2612938  1256648
##  Other - Retired               12101 1020000 807 -1985369  2009572
##  Other - School                 5686 1160000 807 -2265552  2276924
##  Parttime - Retired           690246  418000 807  -130900  1511393
##  Parttime - School            683831  692000 807  -673710  2041372
##  Retired - School              -6416  736000 807 -1451884  1439053
##
## Confidence level used: 0.95
```

Mean comparison with Scheffe's corrections :

```r
Scheffe = cbind(diff - sqrt((K - 1) * qf(1 - alpha, K - 1, n - K)) *
    SE, diff + sqrt((K - 1) * qf(1 - alpha, K - 1, n - K)) * SE)
colnames(Scheffe) = c("Lower", "Upper")
print(Scheffe)
```

```
##                                Lower     Upper
## Fulltime - Housekeeping     -944839.7 1526126.1
## Fulltime - Laid Off         -960890.1 1510075.7
## Housekeeping - Laid Off    -1251533.3 1219432.5
## Fulltime - Not Working      -939561.0 1531404.8
## Housekeeping - Not Working -1230204.2 1240761.6
## Laid Off - Not Working     -1214153.8 1256812.0
## Fulltime - Other            -945612.3 1525353.5
## Housekeeping - Other       -1236255.5 1234710.3
## Laid Off - Other           -1220205.1 1250760.7
## Not Working - Other        -1241534.2 1229431.6
## Fulltime - Parttime        -1623757.3  847208.5
## Housekeeping - Parttime    -1914400.5  556565.3
## Laid Off - Parttime        -1898350.1  572615.7
## Not Working - Parttime     -1919679.2  551286.6
## Other - Parttime           -1913627.9  557337.9
## Fulltime - Retired          -933510.9 1537455.0
## Housekeeping - Retired     -1224154.1 1246811.8
## Laid Off - Retired         -1208103.7 1262862.1
## Not Working - Retired      -1229432.7 1241533.1
## Other - Retired            -1223381.5 1247584.4
```

```
## Parttime - Retired          -545236.5 1925729.3
## Fulltime - School           -939926.4 1531039.4
## Housekeeping - School      -1230569.6 1240396.2
## Laid Off - School          -1214519.2 1256446.6
## Not Working - School       -1235848.3 1235117.5
## Other - School             -1229797.0 1241168.8
## Parttime - School           -551652.0 1919313.8
## Retired - School           -1241898.5 1229067.4
```

```r
confint(pairs(emmeans(fit, ~Work), adjust = "scheffe"))
```

```
##  contrast                    estimate      SE  df lower.CL upper.CL
##  Fulltime - Housekeeping       290643  321000 807  -915274  1496560
##  Fulltime - Laid Off           274593  788000 807 -2687756  3236941
##  Fulltime - Not Working        295922  658000 807 -2177400  2769244
##  Fulltime - Other              289871  962000 807 -3328839  3908580
##  Fulltime - Parttime          -388274  255000 807 -1348368   571819
##  Fulltime - Retired            301972  359000 807 -1049570  1653514
##  Fulltime - School             295556  658000 807 -2177766  2768879
##  Housekeeping - Laid Off       -16050  839000 807 -3171511  3139410
##  Housekeeping - Not Working      5279  718000 807 -2696340  2706898
##  Housekeeping - Other            -773 1000000 807 -3779196  3777651
##  Housekeeping - Parttime      -678918  386000 807 -2129160   771324
##  Housekeeping - Retired         11329  461000 807 -1723055  1745713
##  Housekeeping - School           4913  718000 807 -2696706  2706532
##  Laid Off - Not Working         21329 1020000 807 -3802284  3844942
##  Laid Off - Other               15278 1240000 807 -4632061  4662616
##  Laid Off - Parttime          -662867  816000 807 -3732798  2407064
##  Laid Off - Retired             27379  855000 807 -3186553  3241312
##  Laid Off - School              20964 1020000 807 -3802650  3844577
##  Not Working - Other            -6051 1160000 807 -4358013  4345910
##  Not Working - Parttime       -684196  692000 807 -3285406  1917013
##  Not Working - Retired          6050  736000 807 -2763639  2775739
##  Not Working - School           -365  920000 807 -3458954  3458223
##  Other - Parttime             -678145  986000 807 -4385439  3029149
##  Other - Retired                12101 1020000 807 -3815288  3839491
##  Other - School                  5686 1160000 807 -4346276  4357647
##  Parttime - Retired           690246  418000 807  -883168  2263661
##  Parttime - School            683831  692000 807 -1917378  3285040
##  Retired - School              -6416  736000 807 -2776104  2763273
##
## Confidence level used: 0.95
## Conf-level adjustment: scheffe method with rank 7
```

Mean comparison with Tukeys's corrections :

```r
Tukey = cbind(diff - qtukey(1 - alpha, K, n - K) * SE/sqrt(2), diff +
    qtukey(1 - alpha, K, n - K) * SE/sqrt(2))
colnames(Tukey) = c("Lower", "Upper")
print(Tukey)
```

```
##                                  Lower     Upper
## Fulltime - Housekeeping     -707581.8 1288868.2
## Fulltime - Laid Off         -723632.2 1272817.8
## Housekeeping - Laid Off    -1014275.4  982174.6
## Fulltime - Not Working      -702303.1 1294146.9
## Housekeeping - Not Working  -992946.3 1003503.7
## Laid Off - Not Working      -976895.9 1019554.1
## Fulltime - Other            -708354.4 1288095.6
## Housekeeping - Other        -998997.6  997452.4
## Laid Off - Other            -982947.2 1013502.8
## Not Working - Other        -1004276.3  992173.7
## Fulltime - Parttime        -1386499.4  609950.6
## Housekeeping - Parttime    -1677142.6  319307.4
## Laid Off - Parttime        -1661092.2  335357.8
## Not Working - Parttime     -1682421.3  314028.7
## Other - Parttime           -1676370.0  320080.0
## Fulltime - Retired          -696252.9 1300197.0
## Housekeeping - Retired      -986896.1 1009553.8
## Laid Off - Retired          -970845.8 1025604.2
## Not Working - Retired       -992174.8 1004275.2
## Other - Retired             -986123.5 1010326.4
## Parttime - Retired          -307978.6 1688471.4
## Fulltime - School           -702668.5 1293781.5
## Housekeeping - School       -993311.7 1003138.3
## Laid Off - School           -977261.3 1019188.7
## Not Working - School        -998590.4  997859.6
## Other - School              -992539.1 1003910.9
## Parttime - School           -314394.1 1682055.9
## Retired - School           -1004640.5  991809.4
```

```r
confint(pairs(emmeans(fit, ~Work)))
```

```
##  contrast                   estimate      SE  df lower.CL upper.CL
##  Fulltime - Housekeeping      290643  321000 807  -683694  1264980
##  Fulltime - Laid Off          274593  788000 807 -2118876  2668062
##  Fulltime - Not Working       295922  658000 807 -1702432  2294276
##  Fulltime - Other             289871  962000 807 -2633914  3213656
##  Fulltime - Parttime         -388274  255000 807 -1163995   387446
```

```
##  Fulltime - Retired          301972  359000 807  -790024  1393969
##  Fulltime - School           295556  658000 807 -1702798  2293911
##  Housekeeping - Laid Off     -16050  839000 807 -2565547  2533446
##  Housekeeping - Not Working    5279  718000 807 -2177531  2188088
##  Housekeeping - Other          -773 1000000 807 -3053600  3052055
##  Housekeeping - Parttime    -678918  386000 807 -1850660   492825
##  Housekeeping - Retired       11329  461000 807 -1389990  1412648
##  Housekeeping - School         4913  718000 807 -2177896  2187723
##  Laid Off - Not Working       21329 1020000 807 -3068011  3110669
##  Laid Off - Other             15278 1240000 807 -3739602  3770157
##  Laid Off - Parttime        -662867  816000 807 -3143259  1817525
##  Laid Off - Retired           27379  855000 807 -2569361  2624119
##  Laid Off - School            20964 1020000 807 -3068376  3110303
##  Not Working - Other          -6051 1160000 807 -3522277  3510174
##  Not Working - Parttime     -684196  692000 807 -2785878  1417486
##  Not Working - Retired         6050  736000 807 -2231757  2243857
##  Not Working - School          -365  920000 807 -2794778  2794048
##  Other - Parttime           -678145  986000 807 -3673503  2317213
##  Other - Retired              12101 1020000 807 -3080289  3104492
##  Other - School                5686 1160000 807 -3510540  3521912
##  Parttime - Retired          690246  418000 807  -581015  1961507
##  Parttime - School           683831  692000 807 -1417851  2785513
##  Retired - School             -6416  736000 807 -2244223  2231392
##
## Confidence level used: 0.95
## Conf-level adjustment: tukey method for comparing a family of 8 estimates
```

ii) How do the results of this method compare against the results obtained without the application of the multiple comparisons method?

To begin with manual unadjusted confidence intervals, the Housekeeping - Parttime, Laid Off - Parttime, Not Working - Parttime, Other - Parttime, Parttime - Retired, and Parttime - School pairs are declared statistically significant as they do not include 0 in their intervals, without accounting for the multiple comparisons problem. However, when using emmeans to test pairs with adjust = "none", still without accounting for the multiple comparisons problem, none of the pairs are declared statistically significant so we do not have statistically significant evidence that any pair of Work status group differs in average income.

Scheffe's method is very conservative resulting in none of the differences in average income being declared statistically significant since all intervals include 0 so the aforementioned significant differences are not significant after controlling the FWER and we do not have significant evidence that the average income differs between any pair of the Work status groups at the 95% global confidence level. We can also see how the Scheffe method is very conservative as the confidence intervals are much wider compared to the unadjusted confidence intervals.

Tukey's Honestly Significant Difference method is typically more powerful resulting in more statistically significant pairs than Scheffe which we can see with the narrower confidence intervals but still there are no differences in average income between Work groups being declared statistically

significant while correcting the multiple comparisons problem.

When we compare against the results obtained without the application of the multiple comparisons method using the emmeans output, the conclusion does not change as all methods fail to find any significant differences in average income between any of the Work groups. Using the manual adjustments, some intervals appear significant before correction but none are significant after the Scheffe or Tukey correction.

Step 13:

i) Fit a multiple linear model with all available quantitative and qualitative covariates.

ii) Apply global testing methods to determine whether at least one statistically significant effect exists on your response variable.

iii)

How do the results of these methods compare against each other?

i) Fit a multiple linear model with all available quantitative and qualitative covariates.

```r
global_model = lm(log(Income) ~ Age + Educ + Hours + log(TVHours + 1) +
    MEduc + FEduc + log(Sibs + 1) + Sex + Happy + Marital + Work + Zodiac,
    data = gss)
summary(global_model)
```

```
##
## Call:
## lm(formula = log(Income) ~ Age + Educ + Hours + log(TVHours +
##     1) + MEduc + FEduc + log(Sibs + 1) + Sex + Happy + Marital +
##     Work + Zodiac, data = gss)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.6709 -0.4343 -0.0165  0.3435  6.3160
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)           8.4640010  0.4207527  20.116  < 2e-16 ***
## Age                   0.0108182  0.0036517   2.962 0.003144 **
## Educ                  0.0769796  0.0168501   4.568 5.71e-06 ***
## Hours                 0.0005651  0.0032364   0.175 0.861433
```

```
## log(TVHours + 1)     -0.1502070  0.0848309  -1.771 0.077006 .
## MEduc                -0.0120102  0.0152387  -0.788 0.430853
## FEduc                 0.0216555  0.0132830   1.630 0.103438
## log(Sibs + 1)         0.1401365  0.0714811   1.960 0.050296 .
## SexMale               0.1106975  0.0806101   1.373 0.170070
## HappyPretty Happy     0.3638839  0.1394687   2.609 0.009252 **
## HappyVery Happy       0.5062362  0.1497417   3.381 0.000759 ***
## MaritalMarried        0.4305570  0.1136355   3.789 0.000163 ***
## MaritalNever Married -0.0250756  0.1292355  -0.194 0.846202
## MaritalSeparated     -0.3773618  0.2337053  -1.615 0.106781
## MaritalWidowed       -0.1441457  0.2153231  -0.669 0.503413
## WorkHousekeeping     -0.5044800  0.1627001  -3.101 0.002000 **
## WorkLaid Off          0.1850680  0.3659913   0.506 0.613236
## WorkNot Working      -0.5533673  0.3058133  -1.809 0.070758 .
## WorkOther            -0.0601960  0.4502273  -0.134 0.893673
## WorkParttime         -0.0586110  0.1382797  -0.424 0.671785
## WorkRetired          -0.7676140  0.1993717  -3.850 0.000128 ***
## WorkSchool           -0.1967547  0.3102141  -0.634 0.526100
## ZodiacAries           0.0250841  0.1993702   0.126 0.899909
## ZodiacCancer         -0.0470338  0.1810447  -0.260 0.795093
## ZodiacCapricorn      -0.2112953  0.1794055  -1.178 0.239253
## ZodiacGemini          0.0234715  0.1875985   0.125 0.900464
## ZodiacLeo            -0.2167984  0.1854433  -1.169 0.242727
## ZodiacLibra          -0.2662682  0.1786067  -1.491 0.136415
## ZodiacPisces         -0.1459370  0.1908490  -0.765 0.444697
## ZodiacSagittarius     0.0027182  0.1894261   0.014 0.988555
## ZodiacScorpio        -0.1659318  0.1876191  -0.884 0.376748
## ZodiacTaurus          0.0299261  0.1883216   0.159 0.873781
## ZodiacVirgo           0.1938738  0.1793853   1.081 0.280134
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.071 on 782 degrees of freedom
## Multiple R-squared:  0.186,  Adjusted R-squared:  0.1527
## F-statistic: 5.584 on 32 and 782 DF,  p-value: < 2.2e-16
```

ii) Apply global testing methods to determine whether at least one statistically significant effect exists on your response variable.

H_0 = all the individual null hypotheses are simultaneously true so none of the covariates have a statistically significant effect on the response variable log(Income) H_1 = at least one of them is false so at least one of the covariates has a statistically significant effect on log(Income)

```r
pvalues = summary(global_model)$coefficients[-1, 4]
m = length(pvalues)
alpha = 0.05
```

Bonferroni Test

```
Bonf = min(pvalues)
print(Bonf < alpha/m)
```

```
## [1] TRUE
```

We let B denote the minimum observed p-value. Since B is less than $\frac{\alpha}{m} = \frac{0.05}{32}$ then we reject the global null hypothesis at global significance level $\alpha = 0.05$.

Fisher's Combination Test

```
Fisher = -2 * sum(log(pvalues))
print(Fisher > qchisq(1 - alpha, 2 * m))
```

```
## [1] TRUE
```

Since the Fisher test statistic F is greater than $\chi^2_{2m;\alpha}$ then we reject the global null hypothesis at global significance level $\alpha = 0.05$.

Simes Test

```
Simes = sum(pvalues[order(pvalues)] <= (1:m) * alpha/m)
print(Simes > 0)
```

```
## [1] TRUE
```

Considering ordered p-values $p_{(1)}, p_{(2)}, ..., p_{(m)}$ since $p_{(j)}$ is less than or equal to $\frac{j*\alpha}{m}$ for at least one index then we reject the global null hypothesis at global significance level $\alpha = 0.05$.

iii) How do the results of these methods compare against each other?

All three tests reject the null that none of the covariates have a statistically significant effect on log(Income). The Bonferroni Test uses the minimum observed p-value and compares it to $\frac{\alpha}{m}$. The Fisher combination test combines evidence across all p-values to make a test statistic and compares it the upper $\alpha = 0.05$ quantile of the χ^2_{2m} distribution. The Simes test orders the p-values and tests each index j. Thus all methods utilize the individual p-values to search for a case where one of the null hypotheses is false, so we can conclude that at least one of the covariates has a statistically significant effect on log(Income).

Step 14:

i) Apply multiple testing methods to determine which effects are statistically significant on your response variable by controlling the FWER or FDR at level 5%.

ii) How do the results of these methods compare against each other?